Journal of Volcanology and Geothermal Research 184 (2009) 31–48

Contents lists available at ScienceDirect

Journal of Volcanology and Geothermal Research

journal homepage: www.elsevier.com/locate/jvolgeores

Gravity structure of Piton de la Fournaise volcano and inferred mass transfer during the 2007 crisis

Lydie-Sarah Gailler [a,*], Jean-François Lénat [a], Mathieu Lambert [a,1], Guillaume Levieux [a,2], Nicolas Villeneuve [b], Jean-Luc Froger [a]

[a] Laboratoire Magmas et Volcans, Observatoire de Physique du Globe de Clermont-Ferrand, Université Blaise Pascal, CNRS, 5, rue Kessler, F-63038 Clermont-Ferrand, France
[b] Centre IRD Réunion, BP 172 97492 Cedex, France

ARTICLE INFO

Article history:
Received 10 April 2008
Accepted 27 January 2009
Available online 5 February 2009

Keywords:
Piton de la Fournaise
gravity anomalies
modelling
seismic tomographies
volcano-tectonic
mass transfer

ABSTRACT

The structure and mass transfers at Piton de la Fournaise volcano are investigated using new on land and offshore measurements. The structures are defined using 2 and 3D modelling. Short-wavelength positive anomalies to the W and NW of the central area are interpreted as shallow piles of thick, dense lava flows which have filled volcano-tectonic depressions, several hundreds of meters deep. This finding provides a new insight into the evolution of the volcano. Conversely, the currently active central zone is associated with a negative short-wavelength anomaly, explained by the presence of thin, highly vesiculated and fractured lava flows building the Central Cone. Low density formations underlying the Rivières des Remparts and Langevin areas are tentatively interpreted as breccias related to erosion and/or to southward directed mass-wasting. Topographic offshore highs in the continuation of the SE and NE rift zones of Piton de la Fournaise are characterized by negative gravity anomalies and reversed magnetic anomalies. Their interpretation as hyaloclastites and pillow lavas provides a coherent explanation for both types of signals if we admit that they were primarily constructed during the Alizés stage. Two main deeper, dense structures are also found: the Grand Brûlé complex, and a complex beneath the Plaine des Sables and the Enclos Fouqué. Both are identified as hypovolcanic intrusive complexes. The Grand Brûlé complex (10×18 km in plan view) shows a marked NS-elongated geometry, unlike the other comparable complexes on the island. The Plaine des Sables-Enclos Fouqué complex, a more or less circular structure 5 km in diameter, is attributed to the Ancient Shield stage. It is centered on the Enclos fault, suggesting that the paleo-center of the Ancient Shield was located about 1.5 km eastward of the previously proposed location. An apparent contradiction was noted between the seismic and gravity images of the internal structure. In particular, a high velocity cylinder beneath the summit was not detected in the gravity models. This contradiction was resolved by considering the relative sensitivity of each method.

Finally, the gravity dataset has allowed us to explore the gravity changes that accompanied the major eruptive and volcano-tectonic crisis of April 2007 (350 m collapse of the summit and emission of $\sim120\times10^6$ m^3 of lava). The change in the mass distribution of the edifice was large enough to be detected in the survey data collected in the central area before and after the crisis. However, the accuracy was too low for the precise assessment of the mass transfers.

© 2009 Elsevier B.V. All rights reserved.

1. Introduction

Piton de la Fournaise, a large and active oceanic volcano, has been extensively studied and monitored for several decades. However, a large part of its evolution and of its internal structure remains inadequately documented. Most of its surface is covered by relatively young lava flows and thus only a small part of its geological history is readily accessible at the surface. It has been proposed that Piton de la Fournaise is a relatively young (0.5 Ma) volcano which has grown on the remnants of a larger one called Les Alizés (Lénat et al., 2001). The existence of the latter is only inferred from indirect geological and geophysical observations since it is now concealed by the Piton de la Fournaise volcano.

The evolution of a volcanic system is generally complex because many volcanic (eruptive and intrusive activity), volcano-tectonic (caldera subsidence, flank collapse) and erosive phenomena are interwover in space and time. Geophysical methods provide images of the internal structure which can subsequently be translated in terms of geological structure and evolution. Here, we use gravity measurements to constrain the internal structure of Piton de la Fournaise.

* Corresponding author.
E-mail address: L.Gailler@opgc.univ-bpclermont.fr (L.-S. Gailler).
[1] Present address: GEOLYS sprl, rue des champs Elysées, 4 5590 Ciney, Belgium.
[2] Present address: Géologie des systèmes volcaniques, Institut de physique du globe de Paris, 4, place Jussieu-Case 89-75252, Cedex 05 Paris, France.

0377-0273/$ – see front matter © 2009 Elsevier B.V. All rights reserved.
doi:10.1016/j.jvolgeores.2009.01.024

We have compiled a new gravity map of Reunion Island using recent data collected on land and offshore. In this work we focus on the Piton de la Fournaise volcano. The complete Bouguer anomaly map for the entire Reunion Island and its submarine flanks will be presented and discussed in a separate paper. The new data are used for two different purposes:

(1) Firstly, to study the internal structure of Piton de la Fournaise in more detail than the previously published gravity works. The new data coverage allows us to define clearly the shallow and deep density distribution within the edifice. 3D and 2D 3/4 models have been calculated to assess the geometry and the density of the different structures. A comparison has been made with the seismic velocity models from published tomographies. Discrepancies appear between the two methods and we show that each one has strong and weak points. A new model of the internal structure of Piton de la Fournaise is proposed on the basis of our results and discussed in terms of geology.
(2) Secondly, to make an original study of the gravity changes observed in the central area, for which we have data acquired before and after the eruptive and volcano-tectonic crisis of April 2007. Models have been calculated in order to determine if these events were associated with significant mass displacements inside the edifice.

2. Geological and structural setting

Reunion Island is located in the southernmost part of the Mascarene Basin (Indian Ocean), 800 km east of Madagascar (Fig. 1a). Its origin is related to the hot spot that generated the Deccan Trapps during the Cretaceous and a volcanic chain whose younger elements are Mauritius and Reunion Island (Duncan et al., 1989). Rising above a seafloor of Paleocene age, the Reunion edifice is a flattened cone 200 to 240 km in diameter and approximately 7000 m high (Labazuy and Lénat, 1990; Oehler et al., 2007). The emerged part of the volcanic system represents only a small portion (3% by volume) of the edifice (de Voogd et al., 1999).

The island is composed of two volcanic massifs: Piton des Neiges, a dormant volcano whose last eruptions are 12,000–20,000 years old (Gillot and Nativel, 1982; Deniel et al., 1992) and the currently active Piton de la Fournaise (Fig. 1b). Piton de la Fournaise, located on the

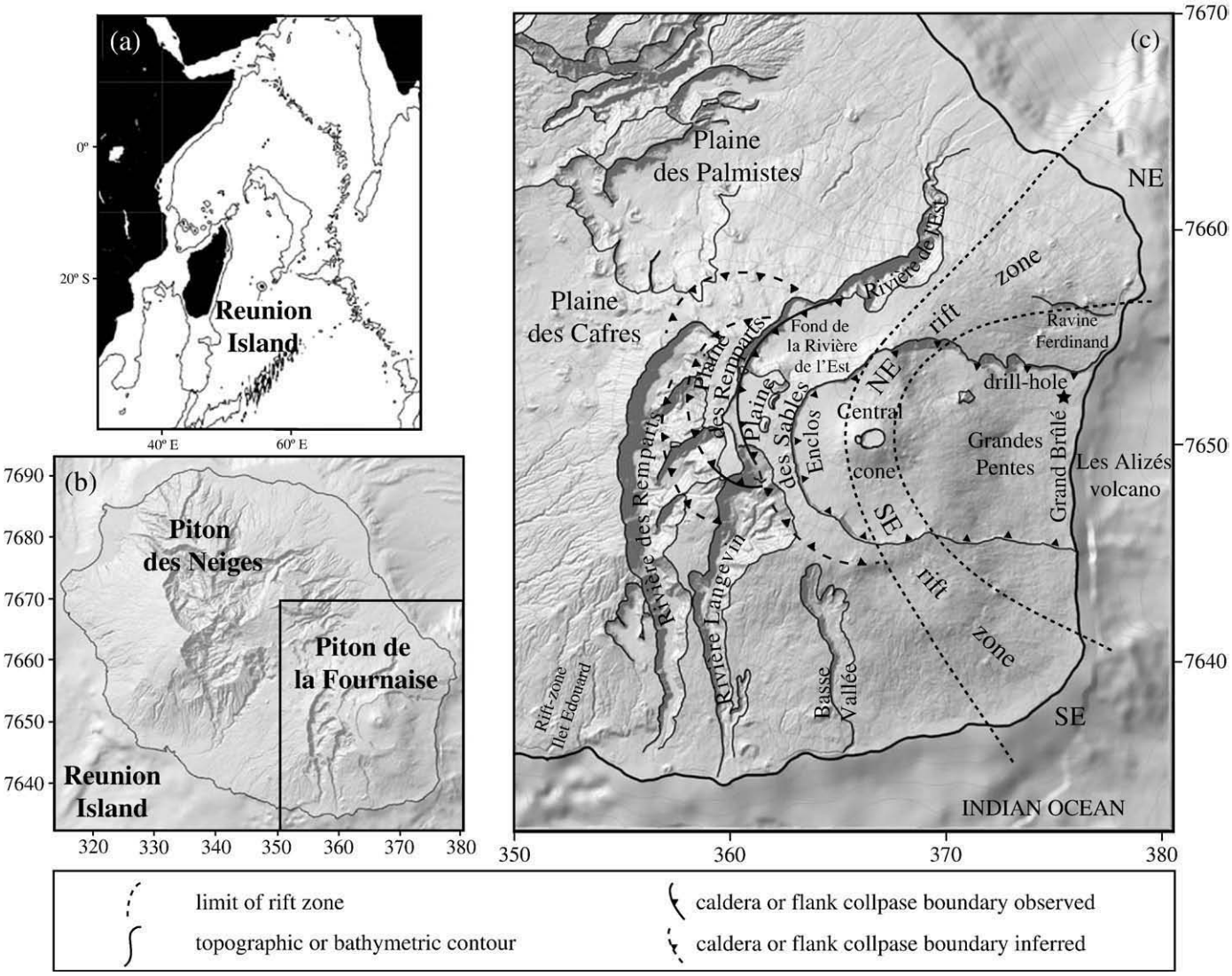

Fig. 1. (a) Location of Reunion Island in the Indian Ocean; coordinates in decimal degrees; (b) shaded relief map of Reunion Island showing the location of Piton des Neiges and Piton de la Fournaise volcanoes; coordinates in kilometers (WGS84, UTM40S). (c) Shaded relief map of Piton de la Fournaise with the main structural features and the location of the main places discussed in the text. Coordinates in kilometers (WGS84, UTM40S).

southeastern part of Reunion Island, is one of the world's most active basaltic shield volcanoes whose activity began more than 500,000 years ago (Gillot and Nativel, 1989). Bachèlery and Mairine (1990) and Bachèlery and Lénat (1993) distinguish two phases of building, the "Ancient Shield" (>0.15 Ma) and the "Recent Shield" (<0.15 Ma), comprising several volcano-tectonic events. Some of these events are recognisable on the present morphology as a series of caldera-like rims and curved valleys (Fig. 1c). For some workers, these structures result from classic caldera collapses (Bachèlery and Mairine, 1990; Bachèlery, 1995). For others, they could be the headwalls of successive eastward moving landslides (Gillot et al., 1994; Labazuy, 1996; Lénat et al., 2001). Merle and Lénat (2003) have developed a hybrid model where a vertical collapse could be generated in response to a flank landslide. Their model could explain the formation of the Enclos Fouqué–Grand Brûlé collapse and possibly that of other structures such as the Plaine des Sables one.

The recent activity has been restricted to mostly effusive eruptions on the Central Cone and along the NE and SE rift zones, with some prehistoric eruptions taking place outside of these main structures. The Central Cone built in the depression of the Enclos Fouqué, is 3 km in diameter and 400 m in height with a mean slope of 15–20°. The summit is occupied by a small pit crater to the west, Bory crater (350 by 200 m), and a larger one to the east, Dolomieu crater (1000 by 700 m; Fig. 1c). Eruptions to the west of the Enclos, in the Plaine des Sables, Fond de la Rivière de l'Est and in the Plaine des Cafres, have ages of a few hundred to a few thousand years (Bachèlery, 1981).

The presence of a third, buried volcano, named Les Alizés, on the eastern coast area of Piton de la Fournaise, has been inferred from gravity (Malengreau et al., 1999), magnetic (Lénat et al., 2001) and drill hole (Rançon et al., 1989) data. Its large intrusive and cumulate complex was encountered at a depth of 1000 m beneath the Grand-Brûlé area by a geothermal exploration drill hole. The complex gives rise to a large gravity anomaly. According to gravity models by Rousset et al. (1989) and Malengreau et al. (1999), it is deeply rooted (5–6 km). Since it cannot be associated with any known volcanic structure at the surface, it has been regarded as an ancient volcanic center pre-dating Piton de la Fournaise. Negative magnetic anomalies offshore from the east coast (Lénat et al., 2001) suggest that rocks belonging to Les Alizés volcano could be present at shallow depth in this area. Alizés activity would pre-date, or be partially contemporaneous, with that of Piton des Neiges.

3. Gravity data

Several gravity land surveys have been carried out since the 1950s on Reunion Island by groups from various French organizations (ORSTOM, BRGM, CNRS—Duclaux et al., 1954; Rechenmann, 1976; Gérard et al., 1980; Demange et al., 1989; Rançon et al., 1989; Rousset et al., 1989), and more recently by Malengreau (1995), Lambert (2003), Levieux (2004). When we started our project, the island was covered with 1634 gravity stations and all available data had been compiled by Levieux (2004). However, this distribution was not ideal to investigate several particularly anomalous areas. Therefore the 2007 survey was carried out to provide a better coverage of selected zones, especially that of Piton de la Fournaise. In addition, the eruption of April 2007, which led to the collapse of the Dolomieu crater, justified the necessity to acquire new data in the active area. Offshore, only a few profiles were available in this area. The ERODER (EROsion et Démantèlement de l'Edifice volcanique de la Réunion) cruise carried out in 2006 provided homogeneous coverage of the submarine flanks from the coast to the base of the edifice on the ocean floor. We have improved the gravity map of Reunion Island using the ERODER survey and about six hundred new data on land (Fig. 2).

3.1. On-land data acquisition and processing

To our knowledge, the surveys from 1979 to 1993 were performed using Lacoste and Romberg gravimeters. The 2003 and 2004 surveys

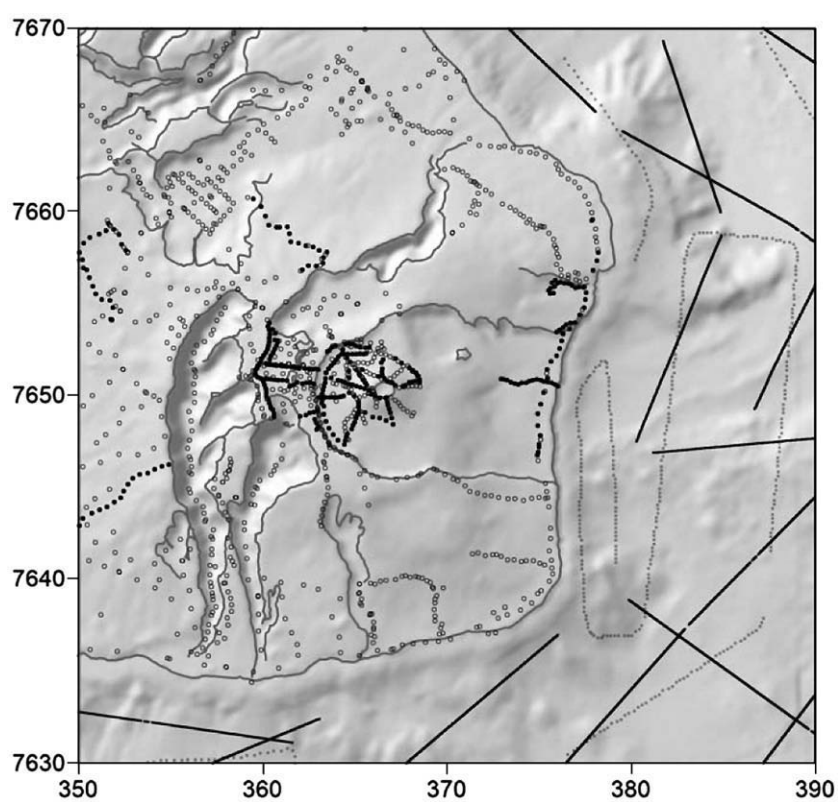

Fig. 2. Data location map of the studied area. ERODER marine survey (2006): black lines; Gallieni marine survey (1995): open circles; Land data from 1979 to 2004: open circles and data from 2007: filled circles. Coordinates in kilometers (WGS84, UTM40S).

were done with Scintrex CG-3M gravimeters and the new May 2007 survey with a Scintrex CG5 gravimeter. Before 1993, the horizontal positioning of the gravity stations was determined from maps and elevations were determined using barometric levelling with a resulting accuracy of approximately ±2 m (Rousset et al., 1989) or geometric levelling (a few millimeters in accuracy). Since 1993, the use of differential GPS techniques to determine station coordinates and altitudes has provided more accurate horizontal positioning and a vertical positioning accuracy of a few centimeters.

The gravity stations are usually distributed along profiles as shown in Fig. 2. Gravity data were tied to the absolute base station established at the Observatoire Volcanologique du Piton de la Fournaise (OVPF): 978638.1143 mGal; 55°5718 E; 21.2086 S; alt 1550 m (M. Amalvic, com. pers. 2004). The observed gravity data are corrected with standard procedures, namely tidal variations and instrumental drift. In a first step, the 1967 International Gravity Formula has been used in order to determine the theoretical gravity at each station location on the Earth's spheroid. Then this value is corrected with the free air gradient, Bouguer slab and terrain corrections. The slab correction has been computed considering the Earth's curvature (Lafehr, 1991b). Finally, the resulting theoretical gravity value is subtracted from the observed one to obtain the Bouguer anomaly.

Terrain corrections have been calculated using Oasis Montaj software from Geosoft. A far zone correction is computed using a coarse regional Digital Elevation Model (DEM) whereas the near zone correction is computed with a more detailed local one. The near zone is a 5 km × 5 km square centered on the station and the far zone extends to 167 km. For the data acquired on land, we have used a 25 m DEM from the Institut Géographique National (IGN) for the near zone and a 500 m DEM for the far zone. For the offshore data, because the measurements were collected further from the surface, we have used a 500 m DEM for the near zone and a 1000 m DEM for the regional corrections. The density of sea water was assumed to be 1.027×10^3 kg m^{-3}.

The large topographic change associated with the 2007 summit collapse was also taken into account in the computation of the terrain corrections for the gravity data acquired after the event. The 25 m DEM used for the data pre-dating the collapse of the Dolomieu crater was modified using the new topographic data from Urai et al. (2007). Terrain corrections vary between 10.14 mGal and 17.64 mGal, with an average of 16.39 mGal for the land gravity stations and between 1×10^4 mGal and 0.18 mGal, with a mean value of 0.05 mGal for the marine stations.

The density correction has been determined from the analysis of the correlation between the Bouguer anomaly and the topography (Nettleton, 1939) along 9 evenly spaced profiles. The density which minimizes the correlation varies along portions of the profiles between about 1.6 and 3.0×10^3 kg m^{-3}. This is due to the presence of anomalies and it is therefore not possible to determine a unique density value which minimizes the correlation between anomaly and topography everywhere on the map. However, a value of 2.2×10^3 kg m^{-3} represents a reasonable average for the covered area. In addition, this is the best density value to minimize the anomaly/topography correlation across the large and deep valleys of Piton de la Fournaise.

For the surveys carried out between 1979 and 2004, the uncertainty on the gravity measurements ranges from 0.066 to 2.80 mGal with a mean value of 1.8 mGal (Lambert, 2003). For the 2007 survey the mean accuracy is about 0.129 mGal for the gravity measurements and 0.05 m for the elevation measurements. The mean accuracy of the complete Bouguer anomaly is about 0.5 mGal with most of the uncertainty linked to the terrain corrections.

3.2. Marine data acquisition and processing

The new marine gravity data come from the 1995 Gallieni survey and from the 2006 ERODER survey. The ERODER cruise (R/V Beautemps-Beaupré) covers the submarine flanks, with a total of 61,106 useful readings acquired around the island and 4918 data in the study area. The distance between measurements along the profiles is 25 m on average. The measurements have been obtained using a KSS31 relative gravimeter from Bodenseewerk and the dataset was tied to a EOST (Ecole et Observatoire des Sciences de la Terre) absolute gravity base located near the harbor (unpublished report from IFRTP—Institut Français pour la Recherche et la Technologie Polaire). Given a negligible drift of the gravimeter (about 0.01 mGal) during the survey no drift correction was applied, but an intersections adjustment between the profiles is required. 21 intersections exist on the 2006 ERODER survey. The histogram of the differences at these crossing points is centred on 0, with a mean value of 0.1558 mGal and a standard deviation of 0.465 mGal. The Gallieni survey has only two intersections with values of −2.75 mGal and 0.98 mGal. Merging the two datasets generates 27 intersections with a mean value of −0.22 mGal and a standard deviation of 1.76 mGal for the differences at crossing lines. These statistics account for the good quality of the marine data. The profiles have been then adjusted using the levelling method of Geosoft Oasis Montaj software.

The marine Bouguer anomaly was also calculated for a terrain density of 2.2×10^3 kg m^{-3}, although submarine vesicular rocks should have a higher density than their dry counterparts because of the effect of water saturation (Kauahikaua et al., 2000). Keeping a homogeneous reduction density for the whole map was preferred in order to make the model interpretation more straightforward.

4. The new Bouguer anomaly map: Qualitative description

The new map of Piton de la Fournaise (Fig. 3) is constructed with significantly better data coverage than the last published one by Malengreau et al. (1999), especially in the Enclos Fouqué, Plaine des Sables and Grand Brûlé areas. We will focus on a 1600 km^2 area centred on the active zone. In this area, 1943 land and 5344 offshore data are available, including 301 new May 2007 data, 4918 from ERODER cruise and 428 from the Gallieni survey. The new data, including those of Lambert (2003), Levieux (2004), the ERODER marine cruise and that of our 2007 survey, have an overall higher accuracy than the previous ones, because they were carried out with modern gravimeters and GPS positioning.

Rousset et al. (1989) and Lesquer (1990) made the first interpretations of the gravity anomalies of the island. Their works show two major positive anomalies associated with intrusive complexes, one located in the central area of Piton des Neiges and the other in the eastern part of Piton de la Fournaise, in the Grand Brûlé area. There is no similar anomaly in the summit area of Piton de la Fournaise, although low amplitude and short-wavelength anomalies were attributed to preferential intrusive zones. The most recent interpretation by Malengreau et al. (1999) provides additional information on a gravity ridge between Piton des Neiges and Piton de la Fournaise and attributes the smaller anomalies over Piton de la Fournaise to lesser bodies of dense rocks emplaced during the early stages of the volcano. The main features and anomalies presented here remain the same as in the Malengreau et al. (1999) map, but their shapes are now better defined and more short-wavelength information is present. These characteristics will allow us to construct more reliable models. We will first describe the main anomalies observed on the new map before turning to the results of the quantitative interpretation of the map (Fig. 4).

4.1. The Grand Brûlé complex anomaly and the rift-zones anomalies

The large positive anomaly of the Grand Brûlé (Fig. 4) has been long recognised (Rousset et al., 1987; Rançon et al., 1989; Rousset et al., 1989) and is clearly associated with a gabbroic and cumulate complex which was encountered by a deep geothermal exploration drill hole at a depth of 1000 m and continued to the bottom of the drill hole at a

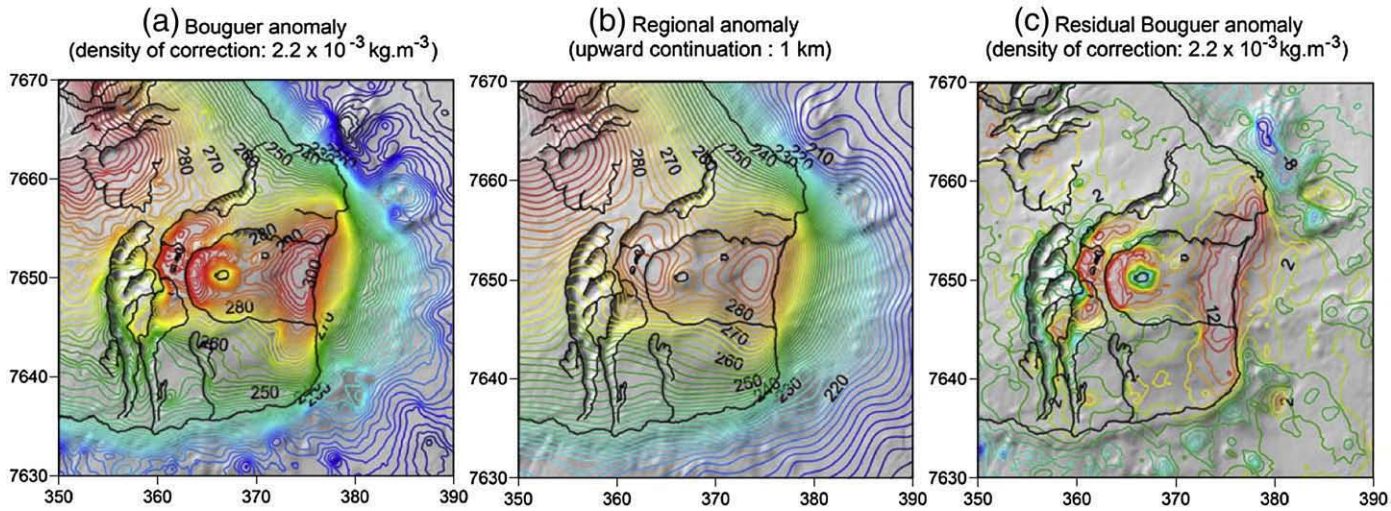

Fig. 3. (a) Bouguer anomaly map computed for a density correction of 2.2×10^3 kg m^{-3}; (b) regional anomaly obtained with a 1 km upward continuation; (c) residual gravity map obtained by subtracting the regional anomaly from the observed one; this residual anomaly is built to extract the short-wavelength signal. Coordinates in kilometers (WGS84, UTM40S). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

depth of 3000 m. This structure was interpreted as the central hypovolcanic intrusive complex of a now concealed volcano named les Alizés (Rançon et al., 1989; Malengreau, 1995; Lénat et al., 2001). The new on land and marine gravity data better constrain the shape and extent of this anomaly, both on the emerged and immerged parts of the edifice. The high intensity of the anomaly accounts for a high density of the source (more than 3.0×10^3 kg m^{-3}) in agreement with that of the rocks observed in the drill hole, together with its large volume and large vertical extent. The complex is elongated in a north–south direction (15–18 km) in plan view, with its NS axis located 1.5 km inland. The marine extension is now well constrained with the new offshore data.

To the N–NE and S–SE of the positive Grand Brûlé anomaly, negative anomalies coincide with offshore constructions in the continuation of the NE and SE rift zones of Piton de la Fournaise (Fig. 4). The origin of these bathymetric highs has been discussed by

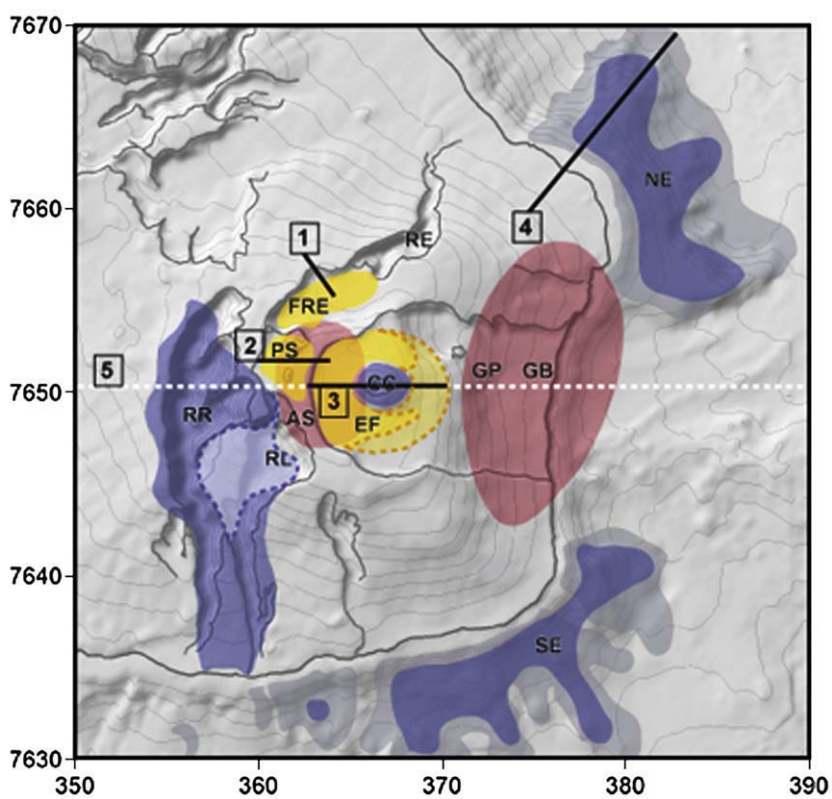

Fig. 4. Interpretive distribution of the shallow anomaly sources (Fig. 3c): Fond de la Rivière de l'Est (FRE), Rivière des Remparts (RR), Rivière de l'Est (RE), Plaine des Sables (PS), Enclos Fouqué (EF), Central Cone (CC), Rift zone Nord Est (NE), Rift zone Sud Est (SE), and deep anomalous bodies: Grand Brûlé (GB) and Ancient Shield of Piton de la Fournaise (AS). Blue areas: short-wavelength negative anomalies; yellow areas: short-wavelength positive anomalies; yellow dashed lines: inferred continuation of short wavelength positive gravity anomalies; red areas: long-wavelength positive anomalies. Offshore topographic highs interpreted by Oehler et al., (2007) as volcanic constructions are shown in grey. Coordinates in kilometers (WGS84, UTM40S). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Lénat et al. (1990), Labazuy (1991), and Oehler et al. (2007). They have been regarded as remnants of old volcanic constructions unrelated to the growth of the recent Piton de la Fournaise (~0.5 Ma). They are associated with reverse magnetic anomalies (Lénat et al., 2001) indicating that they have a coherent magnetization and an age of at least 0.78 Ma. It may thus appear confusing that marine volcanic constructions are also associated with negative gravity anomalies.

4.2. The Piton de la Fournaise anomalies

With no large positive anomaly under the present summit area, the gravimetric signature of Piton de la Fournaise differs greatly from those of the Piton des Neiges and Les Alizés volcanoes. The new surveys since 2001 define more precisely the anomalies on the Piton de la Fournaise massif (Fig. 4). The different features are described bellow from west to east.

The area of the Rivière des Remparts corresponds to a broad, relatively negative anomaly (Fig. 4), which indicates the presence of underlying formations with a density less than the correction density of 2.2×10^3 kg m^{-3}. The explanation of this low density is not obvious from the geology, because the rocks exposed in the rims of the valleys (Bachèlery and Mairine, 1990) do not differ, from a gravity point of view, from the rocks exposed in the other valleys.

To the east of Rivière des Remparts, we observe a curvilinear, positive short-wavelength anomaly over the Plaine des Sables and the Fond de la Rivière de l'Est. This short-wavelength feature was not defined on the previously published gravity maps because of insufficient data coverage. The fact that this anomaly is confined within well defined geological structures suggests the presence of shallow formations with a density significantly higher than the correction density of 2.2×10^3 kg m^{-3}. The subsurface nature of the rocks beneath Plaine des Sables and Fond de la Rivière de l'Est can be inferred from observations in the rims to the north and south of these structures. Between Plaine des Sables and Fond de la Rivière de l'Est, a high escarpment (Plateau des Basaltes) exhibits a pile of thick, horizontal lava flows. This structure is about 130 m high to the bottom of the Plaine des Sables area and about 600 m high to the bottom of the Fond de la Rivière de l'Est (Fig. 5). A similar accumulation is also observed in the 730 m-high scarp between Fond de la Rivière de l'Est and the valley of the Rivière de l'Est. These horizontal lava flows are characteristic of the filling of more or less closed depressions such as caldera floors or craters. Their density is higher than that of the generally thinner lava flows emplaced on the volcano slopes, because of their higher proportion of massive core. Hence, we can infer that the subsurface of Plaine des Sables and Fond de la Rivière de l'Est is composed of several hundred meters of thick, horizontal lava flows.

A long-wavelength positive anomaly is also present to the west of the present summit of Piton de la Fournaise. The high intensity response suggests the presence of a buried dense body by analogy with the dense hypovolcanic complex previously described in the Grand Brûlé area (Rousset et al., 1989; Malengreau et al., 1999). It coincides with the previous central area of Piton de la Fournaise, before it migrated eastwards about 0.15 Ma ago (Bachèlery and Mairine, 1990; Bachèlery and Lénat, 1993). According to Malengreau et al.(1999), the gravity signal could be explained by the presence of a small intrusive complex beneath the Ancient Shield of Piton de la Fournaise. This interpretation is also supported by the occurrence of gabbro and cumulate in this area (Bachèlery, 1981). The new complete dataset available in the Plaine des Sables-Enclos Fouqué area will allow the shape and extent of this major structure to be viewed within the evolution of the Piton de la Fournaise.

Another important pattern of curvilinear short-wavelength anomalies appears around the Central Cone, forming a positive ring around the negative anomaly associated with the cone itself. We do not have direct geological observations to assess the nature of the source of these anomalies. An explanation similar to that proposed above for the short-wavelength anomalies of Plaine des Sables and Fond de la Rivière de l'Est may be conjectured. It implies the formation of thick lava flows in the Enclos Fouqué floor. Accordingly, the short-wavelength positive anomalies could delineate previous depressions filled by thick lava flows. Conversely, the lava flows emplaced on the cone slopes are usually thin, highly scoriaceous, as can be observed in the rim of the Dolomieu crater. Macroscopically, the pile of lava flows that built the Central Cone must have a low density. An exception is a pile of thicker lava flows filling a paleo-pit crater located beneath the western part of Dolomieu crater and the Bory crater. However, the lava flows from this local structure are significantly thinner than the ones exposed in the Plaine des Sables-Fond de la Rivière de l'Est area. Most are only a few meters thick, and the largest are at most 5–10 m thick, whereas the Plaine des Sables-Fond de la Rivière de l'Est are commonly 30–40 m thick. There is also a large proportion of tephra and scoria between the lava flows. Thus, we can infer that, although this local structure should have a slightly higher density than the rest of the cone, it will not change significantly the overall density of the Central Cone. These

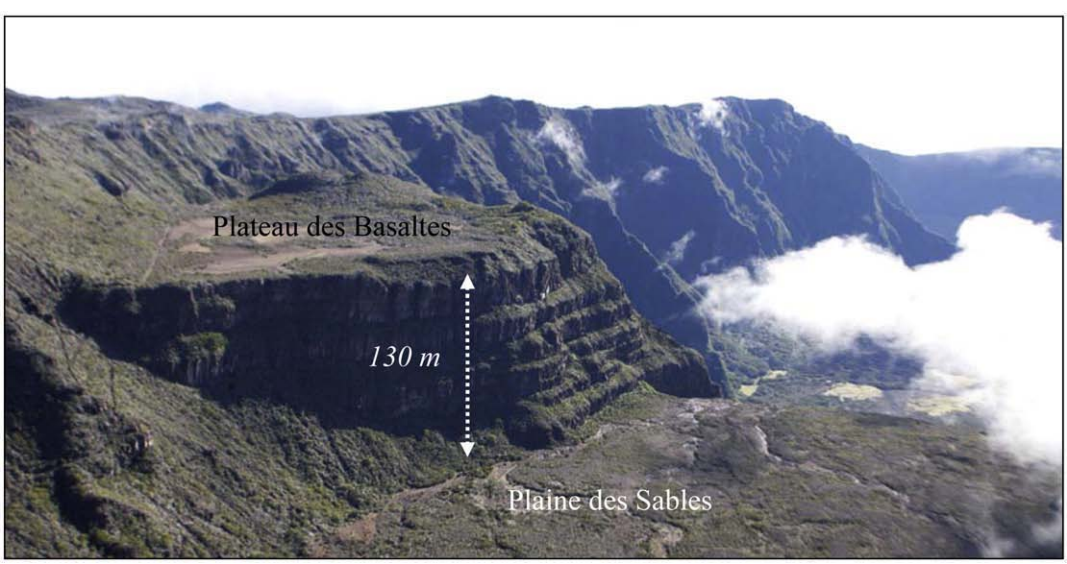

Fig. 5. Plateau des Basaltes in Plaine des Sables-Fond de la Rivière de l'Est area. Built by massive, horizontal thick lava flows it evidences an infilling stage of the Plaine des Sables caldera.

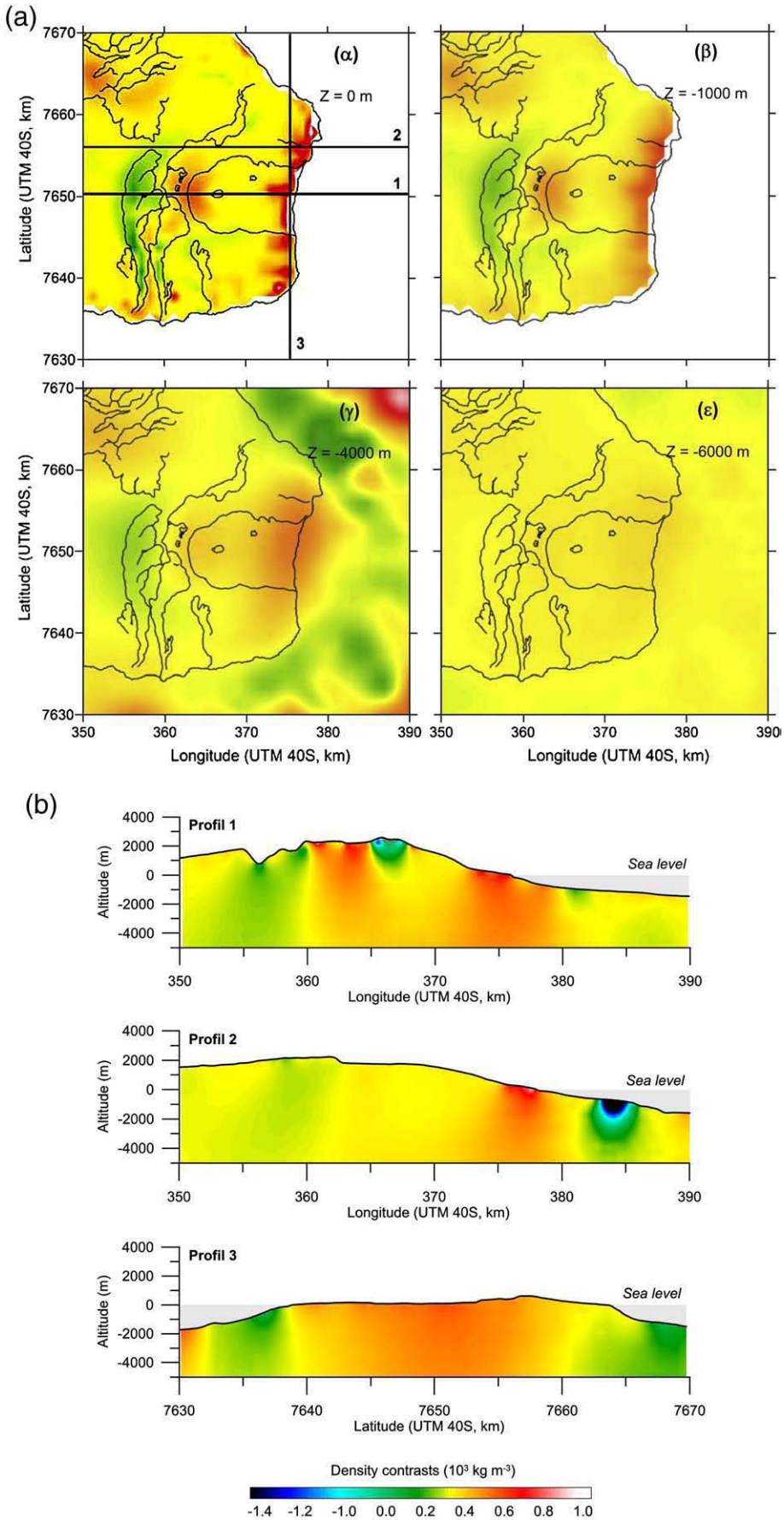

Fig. 6. (a) Horizontal slices in the final model resulting from the UBC-GRAV3D inversion for a correction density of 2.2×10^3 kg m^{-3} at: (α) sea level; (β) 1000 m bsl; (γ) 4000 m bsl; (ε) 6000 m bsl; (b)vertical slices of the same model from profiles shown in panel a. Coordinates in kilometers (WGS84, UTM40S).

observations are coherent with the presence of a negative anomaly above the Central Cone.

5. Gravity modelling and discussion of the structure of Piton de la Fournaise

We have carried out different modelling approaches to study the internal gravity structure of Piton de la Fournaise. Firstly, we have made a 3D inversion of the whole map using the University of British Columbia (UBC) software GRAV3D developed by Li and Oldenburg (1998). Secondly, we have carried out 2D 3/4 modelling using geological and geophysical constraints in order to refine the models for particular zones.

5.1. 3-D model of the central zone of Piton de la Fournaise

The inverted data are extracted anomalies after standard reduction and regional removal are applied. The regional trend is a 2nd degree polynomial fit of the observed data. The data need to be continued upwards because of the strong topography in this area. In our case, the data are continued upwards 125 m above the surface. The subsurface volume is divided into an array of parallelepiped elements whose size is chosen according to the area of interest and to the required resolution. UBC-GRAV3D enables a core region to be defined, beneath the area where observed data exist, and over a surrounding outer region. The latter is necessary to limit boundary effects, especially when the anomalies extend to near or beyond the core region. We have designed a volume with its top just above the highest elevation point. The horizontal mesh size is 1000 m in the outer zone and 250 m in the core region. The elements are 125 m thick near the surface, and increase to 250 m thick at the base of the model at a depth of 20,000 m.

No starting model has been set, but lower and upper density bounds have been imposed (from -2.0 to 2.0×10^3 kg m^{-3}) for each cell. According to these parameters, the 3D density model is composed of $166\cdot166\cdot120$ cells.

Horizontal slices and vertical sections in the 3D model are shown in Fig. 6. Since no constraints have been set for the boundaries of the different structures, the output of the inversion shows smooth structures. We also note an apparent continuity between shallow and deep bodies. Although this could sometimes be real, we show below that in most cases the shallow and deeper structures are not connected. On the basis of this general 3D model, we have chosen to investigate the subsurface structure in more detail with 2D 3/4 models, because the geometrical constraints are more easily managed in 2D.

5.2. 2D gravity modelling

In 2D 3/4 models (we used GM-SYS software), the 2D structures are truncated by the user at some distance chosen in both directions along the perpendicular of the 2D section.

5.2.1. Extraction and modelling of the shallow, short-wavelength anomalies

As noted above in the description of Piton de la Fournaise anomalies, short-wavelength gravity features are obvious in our new gravity map. These high frequency signals can be highlighted using proper filtering methods. Here, we have carried out the separation of the high frequency component by first applying a 1 km upward continuation (Fig. 3b) and then subtracting it from the observed map (Fig. 3a). We observe that when continued upward, the data reduce the effects of the small size shallow structures compared to the larger, deep structures. The resulting residual map obtained by subtracting the upward continued map from the original one (Fig. 3c) highlights the major short-wavelength anomalies of the Fond de la Rivière de l'Est, Rivière Langevin and Rivière des Remparts, Plaine des Sables, Enclos Fouqué, Central Cone, Grandes Pentes and Grand Brûlé areas (Fig. 4).

5.2.1.1. The Fond de la Rivière de l'Est, the Plaine des Sables filling and the Enclos Fouqué-Central Cone subsurface structures.

The inferred filling of a previous depression or valley by a pile of thick massive lava flows in the Fond de la Rivière de l'Est area is modelled using a density of 2.8×10^3 kg m^{-3} (density correction for the map: 2.2×10^3 kg m^{-3}). Such a density provides a model in which the lava pile is bounded to the west by a high angle rim and by a lower angle rim at the east (Fig. 7a). The 600–800 m thickness of the pile is comparable to the altitude difference between the surface of the Fond de la Rivière de l'Est and the nearby bottom of the Rivière de l'Est.

The positive short-wavelength in Plaine des Sables is modelled by a dense structure with a similar density than the massive accumulation imaged in the Fond de la Rivière de l'Est area (2.8×10^3 kg m^{-3}). As in the case of Fond de la Rivière de l'Est, the resulting model appears asymmetric, with a higher angle rim bounding the infill to the west (Fig. 7b). The thickness of this lava pile ranges from approximately 600 m to 800 m, which is comparable to the height between the top of the Plateau des Basaltes (Fig. 5) and the surface of the Fond de la Rivière de l'Est, as well as the thickness of the calculated Fond de la Rivière de l'Est filling. However, this structure is wider than the Fond de la Rivière de l'Est filling, and also thinner towards the east. The maximum thinning more or less coincides with a curvilinear axis along which major volcanic cones (Piton Chisny, Demi Piton and Piton Haüy) are built in the Plaine des Sables area.

In both cases of Fond de la Rivière de l'Est and Plaine des Sables infillings, changing the density of the piles would also slightly change the thickness of the models, but not in a large proportion as long as we keep geologically realistic density values. Such models are thus considered to be a plausible image of the subsurface structures of these areas. Figs. 4 and 7 show the lateral and vertical extent of the inferred paleo-depressions. The Plaine des Sables and Fond de la Rivière de l'Est paleo-depressions are clearly bounded to the west by the fault of the Plaine des Sables which collapsed about 40 ka (Gillot and Nativel, 1989). The models suggest that this fault had a vertical offset of nearly 1000 m. However, the base of the infilling beneath the Fond de la Rivière de l'Est is about 500 m lower in elevation than that beneath the Plaine des Sables. This elevation difference suggests that a paleo-Rivière de l'Est had been carved previously to its infilling by lava flows. The presence of horizontal thick lava flows indicates that the flows were trapped in depressions such as calderas, craters or even valleys. The origin of these short-wavelength components as piles of thick dense lava flows is well established in the Fond de la Rivière de l'Est and Plaine des Sables areas where they are observed in cliffs (Fig. 5). The depressions may have been progressively filled by subsequent effusive activity from a center located to the east, as suggested by Bachèlery and Mairine (1990). Therefore the gravity may be used to map paleo-depressions infilled by thick lava flows.

Within the Enclos, the Central Cone coincides with a circular negative anomaly surrounded by a ring of positive anomalies. We do not have outcrops to determine the source of the positive short-wavelength ring around the Central Cone. However, we can reasonably infer a similar origin to that of the Plaine des Sables and Fond de la Rivière de l'Est shallow anomalies, because we also note that the ring is bounded by the Enclos Fouqué fault to the west. To the south and to the north, the data coverage does not allow us to map the outer limit of the ring. To the east the ring is also loosely defined by the available data. Where the anomaly is well defined to the west, the maximum here suggests that the pile of lava flows is the thickest or the densest in this area. The corresponding model (Fig. 7c) suggests that the paleo-depression of the Enclos is filled by an accumulation of massive thick lava flows ranging from about 600 to 800 m in thickness for densities of 2.8×10^3 kg m^{-3} and 2.6×10^3 kg m^{-3} respectively.

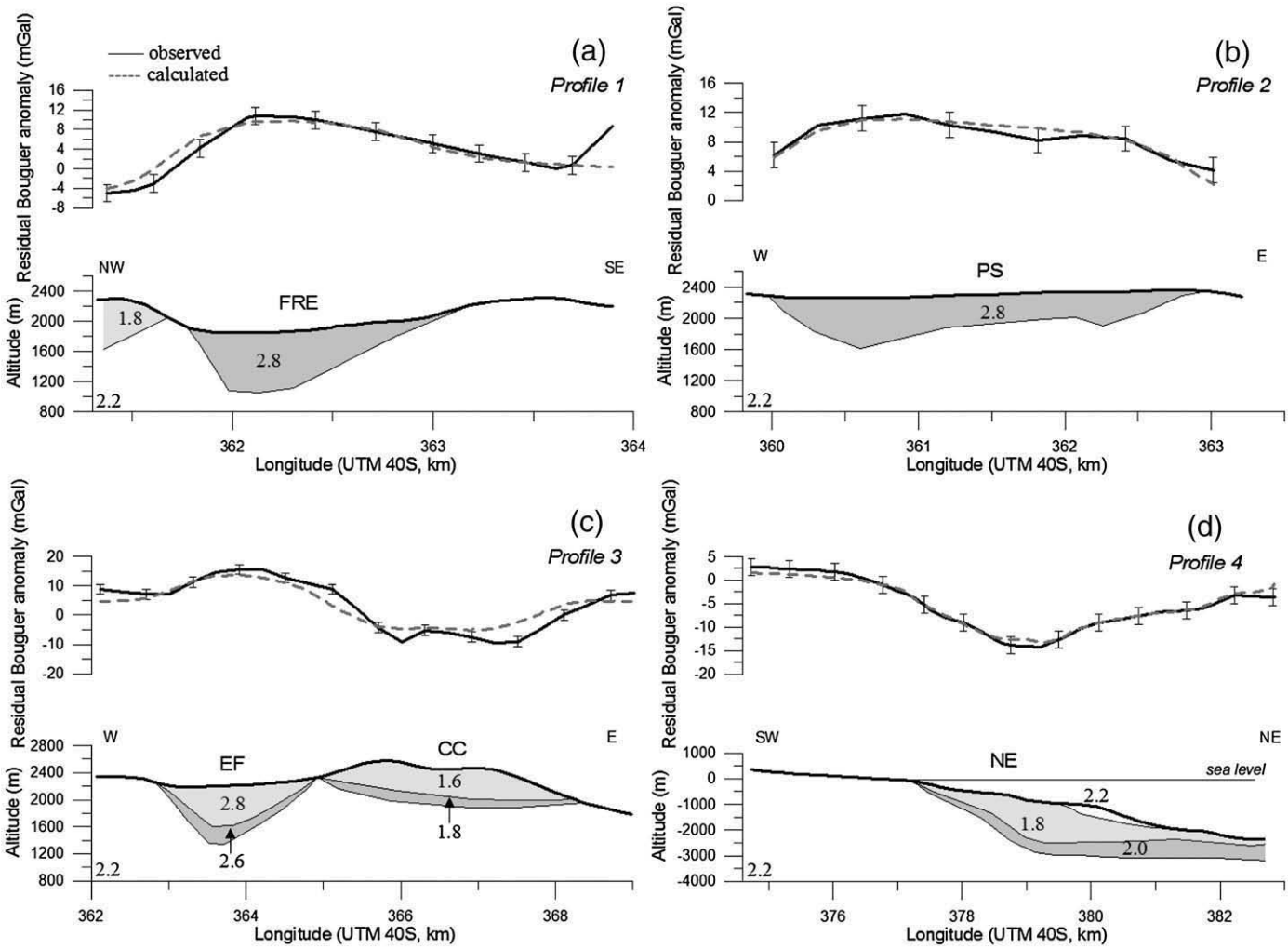

Fig. 7. 2D 1/2 and 2D 3/4 models (profiles are located in Fig. 4) of the short-wavelength anomalies from the residual gravity field (Fig. 3c): (a) profile 1 in Fond de la Rivière de l'Est (FRE) area. The 2D 3/4 structure is limited to a distance of 2 km southward and 5 km northward; (b) profile 2 in Plaine des Sables (PS) area .The 2D 1/2 structure is limited to a distance of 4 km perpendicular to the profile in both directions; (c) profile 3 in Central Cone (CC) and Enclos Fouqué (EF) areas. The 2D 1/2 structure is limited to a distance of 2 km perpendicularly to the profile direction for Central Cone (CC) and of 5 km for Enclos Fouqué (EF); (d) profile 4 in the offshore NE rift zone (NE) area. The 2D 3/4 structure is limited to a distance of 7 km south-eastward and 3 km north-westward from the profile direction. Densities are in 10^3 kg m^{-3}. The observed gravity curve is calculated for a correction density of 2.2×10^3 kg m^{-3}, error bars are shown.

The Central Cone itself is modelled with a very low density structure. For densities of 1.6×10^3 kg m^{-3} and 1.8×10^3 kg m^{-3}, the thickness of this structure is estimated as being 400 and 550 m respectively. The nature of the rocks observed in the rim of Dolomieu crater provides a convincing explanation of the gravity low in this area; the cone is mainly composed of thin, highly vesiculated and fractured lava flows. Clearly, the macroscopic density of the rocks exposed in the rim must be very low, considering the percentage of void space and the high degree of fracturing. Moreover, the existence of a hydrothermal system at very shallow depth (Lénat et al., 2000) may further contribute to decrease the bulk density of this area by hydrothermal alteration. However, the negative gravity signal of this zone appears to contradict the results of seismic tomographies that have detected a shallow high velocity body beneath the summit (Nercessian et al., 1996; Hirn et al., 1999; Brenguier et al., 2007; Prôno et al., 2009-this issue). Nercessian et al. (1996), using a passive tomography approach with local earthquakes, propose the presence of a high-velocity plug (1.5 km in diameter) beneath the central zone, surrounded by a low-velocity ring. This structure has been refined by Hirn et al. (1999) using a passive tomography approach (Fig. 8a). A recent 3D passive tomography (Prôno et al., 2009-this issue), based on the 1998 crisis earthquakes, has provided a high resolution model for the central zone alone (Fig. 8b). The main features of the previous tomographies, i.e. the shallow high-velocity body beneath the summit and its low velocity surrounding, are also imaged, but with better accuracy.

Seismic wave velocity and density are two correlated parameters and various velocity–density relationships have been defined in the literature. Here we have used the formula proposed by Gebrande et al. (1982) which describes the following quantitative relationship between longitudinal wave velocity V_P (in km s^{-1}) and density ρ (in 10^3 kg m^{-3}) for volcanic rocks:

$$V_P = 2.81\rho - 2.37 \pm 0.18.$$

Using this relationship, the internal structure imaged by the seismic methods has been converted into density layers (Fig. 8) and the associated gravity signal has been calculated. Clearly, the theoretical gravity signal of the seismic tomography significantly departs from the observed one in the central area. There is an apparent contradiction between the high-velocity body (and hence, high density structure) imaged in the seismic tomographies, and the lack of a positive gravity anomaly beneath the summit craters, which has to be explained. Both methods are undoubtly reliable, but they may accentuate different structures. The strong gravity signal created by the shallow low density Central Cone will mask a more moderate signal from a denser column

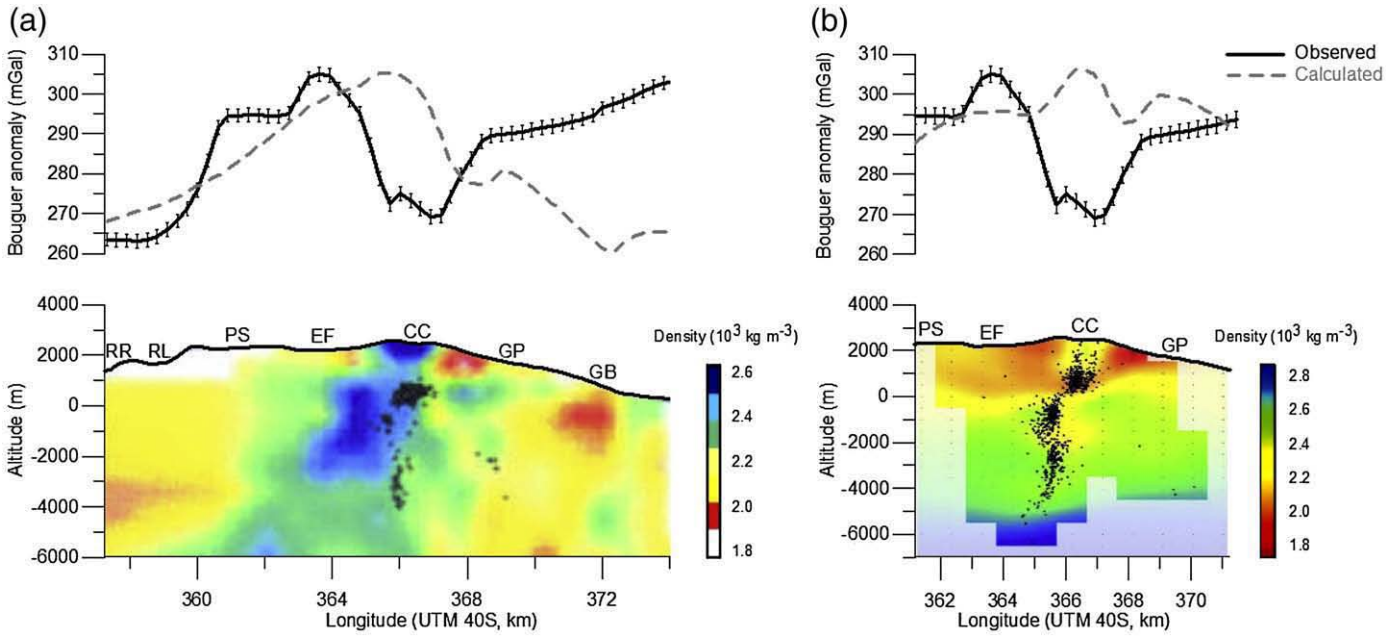

Fig. 8. 2D models of the west to east profile 3 (shown in Fig. 4), on (a) seismic tomography of Hirn et al. (1999); (b) seismic tomography of Prôno et al. (2009-this issue). The seismic velocities have been converted to densities as explained in the text. Labels refer to the main places discussed in text: Rivière des Remparts (RR), Rivière Langevin (RL), Plaine des Sables (PS), Enclos Fouqué (EF), Central Cone (CC), Grandes Pentes (GP), Grand Brûlé (GB).

beneath the summit. Similarly, the ring of dense lava flows inferred from the gravity around the Central Cone may also hide the signal from the low velocity ring of the seismic tomographies. These observations suggest that seismic tomographies are better suited to define deeper structures than subsurface ones, at least with the station and source coverage used in the published tomographies. Conversely, we note that the gravity signal is strongly influenced by shallow structures. It is, therefore, very important to separate the shallow and deeper sources during interpretation of gravity data.

5.2.1.2. Rivière des Remparts; Rivière Langevin.

The negative anomaly imaged in the Rivière des Remparts and Rivière Langevin areas is more difficult to explain, and two major hypotheses can be envisaged. Firstly, the right-bank cliff of Rivière des Remparts is composed of formations belonging to the Ancient Shield of Piton de la Fournaise (Bachèlery and Mairine, 1990). These formations are locally overlain on the left-bank by the sub-horizontal formations of the Recent Shield. During the Ancient Shield period, large episodes of erosion and the collapse of the first caldera (Rivière des Remparts caldera) took place (Bachèlery, 1981; Chevallier and Bachèlery, 1981; Bachèlery and Chevallier, 1982). The present Rivière des Remparts corresponds roughly to the western limit of the Rivière des Remparts caldera. The erosion has excavated a paleo-Rivière des Remparts, which can be reconstructed with the lahar and detrital layers found near the base of the valley rims. At depth, this paleo-valley could be filled by a large proportion of low density erosion material, giving rise to the observed gravity low. Secondly, Oehler et al. (2007) have described large mass-wasting deposits on the south submarine flanks, as an offshore continuation of the Rivière des Remparts and Rivière Langevin. The source area of these deposits could be located in the Rivière des Remparts-Rivière Langevin area, thus low density breccias could be present at depth in this area.

5.2.1.3. The Grandes Pentes–Grand Brûlé subsurface structures.

A shallow dense (2.8×10^3 kg m^{-3}) structure about 300–400 m thick, similar to the dense filling modelled in the Enclos Fouqué area, has been modelled in the Grandes Pentes area (Fig. 9). It would be the eastern part of the dense ring that seems to surround the Central Cone. In this case, however, it is difficult to explain the formation of thick lava flows, because the flows are unconstrained by a topographic barrier in this area. We can therefore suspect that the lack of measurements along the Grandes Pentes area has created artifacts in the data interpolation, rendering the modelling unreliable in this area.

A conspicuous short-wavelength anomaly is observed near the eastern seashore of Piton de la Fournaise. Here, the presence of a pile of thick, dense lava flows can unambiguously be ruled out, because they have not been observed in two drill holes in the Grand Brûlé area (Courteaud, 1996; Rançon et al., 1989). Since this short-wavelength anomaly practically coincides with the extent of the Grand Brûlé complex (described below), it could correspond to the signal at the top of the dense complex, encountered at a depth of only 1000 m in the deep exploration drill hole (Rançon et al., 1989). Note also that the fit of the model is better if the offshore part of Grand Brûlé area is modelled (Fig. 9) with a low density layer (1.8×10^3 kg m^{-3}).

5.2.1.4. The NE and SE offshore structures (Fig. 7d).

As mentioned above, the marine constructions continuing along the NE and SE rift zones coincide with negative gravity and magnetic anomalies. In theory, such magnetized structures should also be characterized by a positive gravity response. The interpretation of these features must therefore be coherent with their physical characteristics. A possible explanation for these offshore low density but magnetized structures would be to consider them as being mainly composed of hyaloclastites. This rock type is often characterized by a lower density than lava flows (Moore, 2001) and by stable thermo-remanent magnetization (Harrison and Ball, 1974) though weaker than that of massive lava flows.

The NE topographic high has been modelled by a stratified structure whose density increases (from 1.8 to 2.0×10^3 kg m^{-3}) from the seafloor down to about 2000 m in depth (Fig. 7d). This density pattern accounts well for the short-wavelength anomaly in this area and is coherent with the observations reported by Moore (2001). From the analysis of drill cores in Hawaii, he distinguishes two parts in a hyaloclatite sequence. The upper one is composed of friable and uncemented hyaloclastites, which are highly permeable and porous. At depth, the hyaloclastites become cemented and less porous. Accordingly, we propose that the NE and SE offshore topographic highs are hyaloclastite deltas.

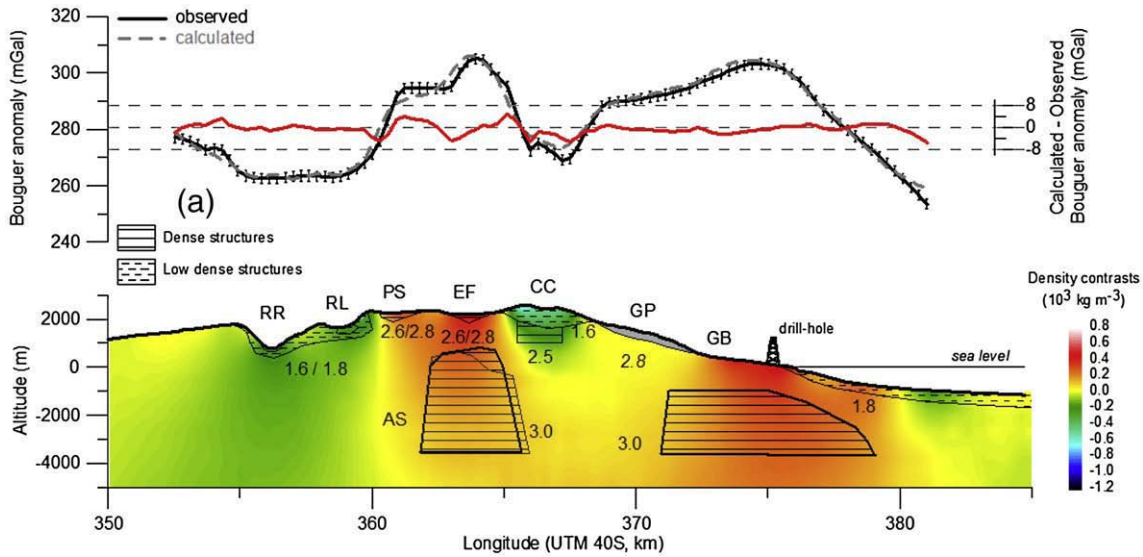

Fig. 9. 2D 1/2 model of the west to east profile 5 (shown in Fig. 4), superimposed on a section of the 3D inversion model. Grand Brûlé deep body is truncated at a distance of 5 km perpendicular to the profile direction; the intrusive complex of the ancient volcanic center of Piton de la Fournaise is truncated at a distance of 10 km perpendicular to the profile direction. The pile of thick, dense lava flows in the Grandes Pentes area appears in grey since the presence of this structure is not well established (see text). The central high velocity body is considered. The two alternative models discussed in the text for the Ancient Shield structure are shown (structure either affected or not by the Enclos Fouqué fault: grey or black structure respectively). The observed gravity curve is calculated for a correction density of 2.2×10^3 kg m^{-3}, error bars are shown. Labels refer to the main places discussed in text: Rivière des Remparts (RR), Rivière Langevin (RL), Plaine des Sables (PS), Enclos Fouqué (EF), Central Cone (CC), Grandes Pentes (GP), Grand Brûlé (GB).

5.2.2. Modelling of the deeper structures

Two main, deeper anomalies are present in the studied area. The largest one is located on the eastern coast and coincides with the hypovolcanic complex attributed to Les Alizés volcano. The second one is a positive anomaly stretching over the western part of the Enclos Fouqué, the Plaine des Sables and the Fond de la Rivière de l'Est. Note also a third anomaly, to the NW of the map, which belongs to Piton des Neiges, and will not be discussed in this work. Constraints from geological observations and geophysical studies have been used for modelling the deeper structures.

The eastern anomaly is well constrained by the data from a deep drill hole. The intrusive layered complex encountered by the drill hole (Fig. 1) in the Grand Brûlé area (from 1000 m to the bottom of the drill hole at 3000 m in depth i.e. about 830 m bsl to 2830 m bsl); (Rançon et al., 1989) is composed successively of sheets of gabbro injected into lavas and hyaloclastites, a zoned intrusion of olivine gabbro and dolerite, and a stratified cumulate gabbro–wehrlite–dunite. According to Demange et al. (1989), the density of this type of intrusive rock generally ranges from 3.1 to 3.35×10^3 kg m^{-3}.

As previously mentioned, the improved coverage of the anomaly, both on land and offshore, allows us to better define the geometry of the source body. We have modelled the Grand Brûlé anomaly with a large dense body, markedly elongated in the NS direction (EW extent of about 10 km and a NS extent of about 18 km), with a density of 3.0×10^3 kg m^{-3}. The bottom of the model is at a depth of 4000 m, but this parameter is not well constrained. However, the data from the drill hole and the amplitude of the anomaly clearly show that the complex is rooted at a depth of several kilometers. To the west, the complex extends approximately to the base of the Grandes Pentes and is limited to the east at less than 5 km offshore. To the north it extends beyond the rim of the Grand Brûlé and beyond a partially buried rim called the Ravine Ferdinand rim (Fig. 1). To the south the complex also spreads beyond the southern rim of the Grand Brûlé. We observe that the topography of its roof is tilted eastward in the model near the sea shore (Fig. 9). It is well known through the works of Labazuy (1996), de Voogd et al. (1999) and Oehler et al. (2007) that the eastern submarine flank is covered by several kilometers of mass-wasting deposits. Oehler et al. (2007) have attributed an extensive surface of debris avalanche deposits at the base of the eastern submarine flank to mass-wasting events affecting Les Alizés volcano. Similarly, Lénat et al. (2001) suggest that only some remnant, reversely magnetized, parts of Les Alizés volcano still exist near the coast, and that the bulk of this volcano has been destroyed by mass-wasting events. Consequently, the top of the complex at the drill hole site could be a glide plane overlain by landslide products as suggested by Courteaud (1996). This interpretation could explain the asymmetrical structure required to account for the gravity response.

The source of the deeper, western anomaly cannot be identified by direct observation, but the occurrence of gabbros and cumulate xenoliths in lava flows and tephras in the area above the anomaly suggests that it is created by a hypovolcanic intrusive complex. Bachèlery (1981) describes frequent gabbroic and peridotitic enclaves within the formations from eruptions (such as that of Piton Chisny) in the Plaine des Remparts, Plaine des Sables and Central Cone areas. These xenoliths suggest that the magmas sample underlying intrusive and cumulate complexes. The distribution of these mafic xenoliths to the west of the Enclos is in agreement with the presence of a volcanic center beneath Plaine des Sables during the Ancient Shield period (Bachèlery and Mairine, 1990). The dense complex is also imaged by the seismic tomography of Hirn et al., (1999) where a high velocity body extending between 1 and 3 km bsl is mapped between the Central Cone and Plaine des Sables areas (Fig. 8a).

Accordingly, we model the anomaly with a dense structure (3.0×10^3 kg m^{-3}) beneath the Plaine des Sables-Enclos Fouqué area. The structure extends from about 700 m asl to 3700 m bsl and has a lateral extent of 2 to 4 km. An important issue for the modelling is to establish whether or not the dense body is affected by the fault that limits the Enclos. If the body was not cut by the fault, it would signify that the Enclos fault is listric, with a shallow dip at depth. We have tested the two geometries (Fig. 9): (1) the top of the body is not perturbed by the fault (i.e. continuous across the fault) and (2), the eastern side of the dense body is offset downwards by the fault. We obtain models that fit well with the observed data for both cases, so the results are not conclusive. The factor that limits the discrimination between the two tectonic patterns is the presence of the strong signal of the shallow component of the gravity anomaly attributed to the

infilling of a paleo-depression in the western part of the Enclos (see above). The response difference between the two models of the deeper structure can be easily compensated for by modifying slightly the geometry of the shallow structure.

6. The April 2007 Dolomieu collapse event

A major collapse of the summit zone occurred during the volcanic and volcano-tectonic crisis of April 2007. Since our dataset comprises measurements that both pre- and post-date this major event, we have studied the gravity signal related to the new mass distribution of the edifice.

6.1. Description of the collapse

A series of historical collapse and replenishment events of the Dolomieu crater have been observed, ranging from the collapse of small (tens of meters in diameter) pit craters in 1953, 1986 and 2002 (Lacroix, 1936; Lacroix, 1938; Bachèlery, 1981; Delorme et al., 1989; Longpré et al., 2007), to the larger collapse of the eastern side of the Dolomieu crater (1931) and even its whole collapse (2007). Since the large 1931 collapse, the Dolomieu crater has been gradually infilled by the products of subsequent eruptions, resulting in its complete replenishment at the end of the August 2006–January 2007 eruption, with lava flows spilling over the rim at its lowest elevation from the vents inside the crater.

The April 2007 collapse of the summit crater was associated with exceptional eruptive activity (Michon et al., 2007; Urai et al., 2007). Following several days of intense volcano-tectonic seismic activity beneath the summit, an intrusive seismic crisis began on March 30, 2007 at 16:25 UTM. A short-lived eruption began at 18:50 from a N120 fissure located at 1900 m asl at the SE base of the Central Cone. The seismic tremor ceased on March 31, marking the end of this 10 hours eruptive phase which emitted less than 10^6 m^3 of magma. However, the summit volcano-tectonic seismicity continued, and a second eruptive phase began on April 2nd at low altitude (about 650 m), 7 km from the summit. The new fissure was located along the continuation of the previous one (Michon et al., 2007). This 26 day-long eruption was exceptional in volume (~100–140 × 10^6 m^3) and output rate (up to 100 m^3 s^{-1}). The lavas, and especially those emitted during the last phase, contained a large percentage of olivine (Famin et al., 2009) suggesting that the eruption had probably drained a shallow reservoir. The seismic activity beneath the summit (between 0 and 500 m in depth), was interpreted as resulting from the fracturing of rocks above the drained reservoir. A summit inward deflation of several centimeters was observed on GPS stations near the rim of the Dolomieu. The seismic signal then changed from continuous to cyclic, with seismic peaks suggesting a step by step collapse of the rock column beneath Dolomieu crater. On April 5th, a paroxysmal seismic phase, characterized by an intensification of the seismicity and the tremor, announced the beginning of the Dolomieu crater collapse (Michon et al., 2007). Fig. 10 shows the Dolomieu crater before and just after its collapse. A fluctuating tremor was recorded until the end of the eruption on May 1st. This eruption was the most voluminous at the Piton de la Fournaise over the last two centuries. According to ground measurements carried out by the Volcanological Observatory, the maximum depth of the collapse was about 320–340 m. The collapse volume, estimated at 100–120 × 10^6 m^3 by Michon et al. (2007), was later confirmed by Urai et al. (2007) using ASTER stereo images.

6.1.1. Gravity changes associated with the April 2007 volcano-tectonic events

The compilation of gravity data acquired at different times is common practice in the construction of gravity anomaly maps. There is no objection for doing so as long as the data have comparable accuracy,

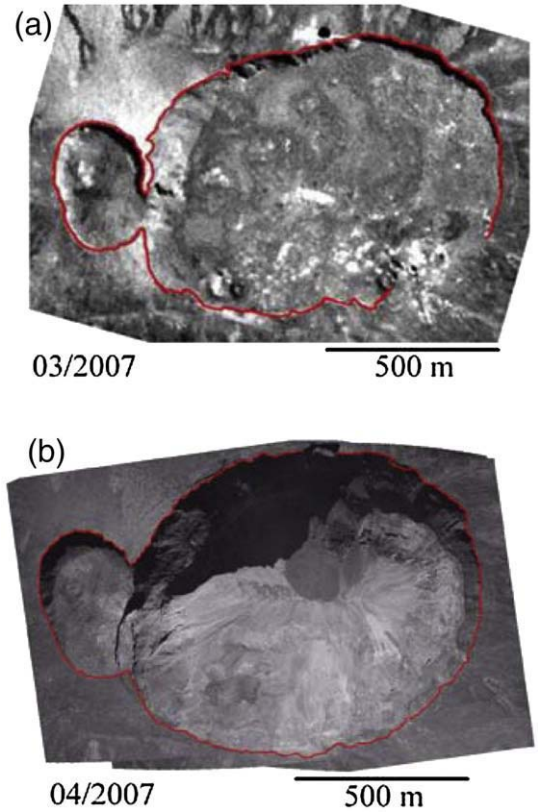

Fig. 10. (a) The Dolomieu crater infilled by successive eruptions just before the April 2007 collapse; (b) the Dolomieu crater just after its April 2007 collapse; depth = 320–340 m, volume = 100–120 × 10^6 m^3 (Michon et al., 2007; Urai et al., 2007). Photography: Nicolas Villeneuve.

and that no large mass transfer has taken place in the concerned area. The April 2007 volcano-tectonic event in the summit zone of Piton de la Fournaise may be a case where mass transfers were large enough to perturb the static gravity field. However, the gravity monitoring of volcanoes to track mass transfers at depth is usually performed using techniques that significantly differ from the technique of gravity surveying aimed at studying the static structure of volcanoes (Rymer, 1989; Brown et al., 1991). The gravity survey data used in this study do not have the accuracy required for monitoring purposes, both in the elevation and the gravity measurement precision. For example, the pre-April 2007 data in the central zone were produced by barometric levelling (Rousset et al., 1989), with an accuracy estimated at approximately 2 m, while an accuracy of a few centimeters is generally required for gravity monitoring. However, as we show below, the April 2007 crisis has created mass transfers large enough to be detected in the survey data in the central area.

The pre- and post-April 2007 data coverage (Fig. 11) is sufficient to consider that the comparison of the two datasets is significant, although the stations do not exactly coincide. The comparison has been carried out using free air anomaly because the map has the advantage of being free from the errors associated with terrain corrections of the Bouguer anomaly.

Several profiles have been selected in areas where gravity data pre- and post-dating the collapse are available (Fig. 12). Profile comparisons have been carried out in areas both far from the summit and in the central zone. For the remote areas, the data coverage allows us to compare the pre- and post- collapse data in the Plaine des Sables and Grand Brûlé areas. Within the Plaine des Sables area (profile AB on Fig. 12a), the 2007 data are denser than previously. Post-April 2007 data values are slightly lower, but the difference remains within the estimated error with a mean difference of −0.8 mGal, considering

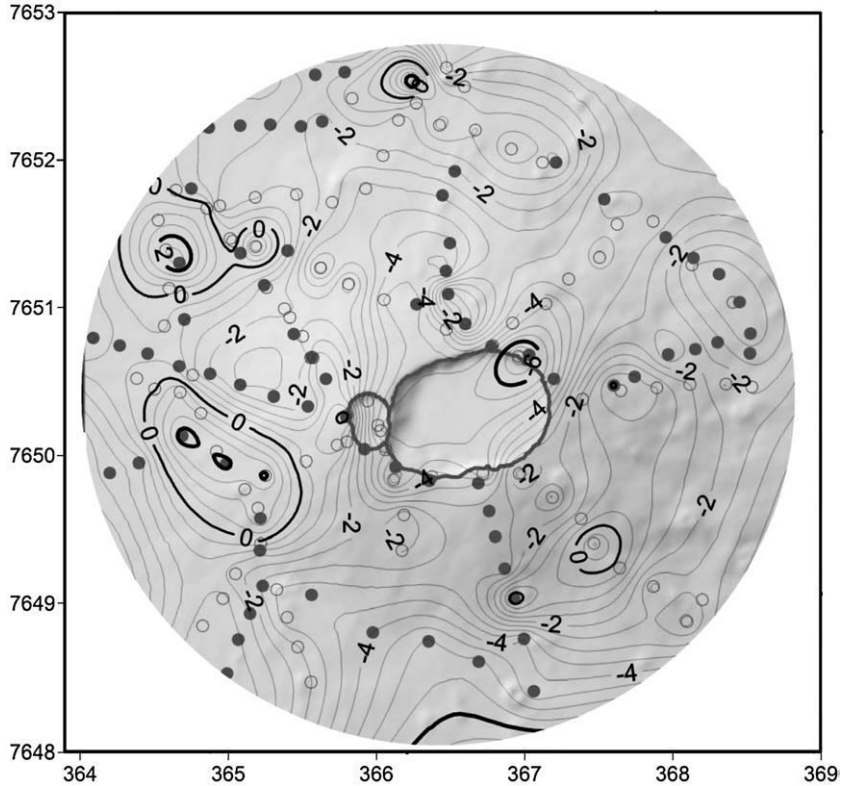

Fig. 11. Contours of the free air anomaly difference between pre- and post-April 2007 Dolomieu collapse (contour interval: 0.5 mGal). Relative maximum (−6 mGal), minimum (2 mGal) and 0 values are shown with thicker contours. Open circles: pre-collapse data; filled circles: post-collapse data. Anomaly in 10^3 kg m^{-3}; Coordinates in kilometers (WGS84, UTM40S).

the data density and the conditions of acquisition (instruments and levelling) between the two sets of measurements. In the Grand Brûlé area, the difference is of the same order as in the Plaine des Sables. Conversely, near the summit (profile CD on Fig. 12b), we observe significant differences in the free air anomaly. The west to east profile shows a mean difference of −0.8 mGal and a maximum observed difference of −4.5 mGal for the free air anomaly. Therefore, the comparison shows that no changes greater than the inferred error are detected outside of the central area, but that near the summit the changes are significant. The difference between the two datasets shows a noise that can be mostly inferred to the difference in accuracy of the two datasets. For that reason, the difference profile in the central area has been smoothed using a low pass median filter. It will be analysed in the next section.

6.2. Analysis and interpretation of the related gravity changes

6.2.1. Possible causes of the observed gravity variations

Three types of phenomena may be suspected to have generated free air anomaly variations (Fig. 13a). The first one is obvious and directly related to the topography change. In the collapsed crater, air has replaced the rocks (Fig. 13b). The two other possible phenomena are more conjectural and derive from geological considerations. Firstly, we can infer that the collapse has generated rock fracturing in the column beneath the crater, therefore decreasing the density of the column (Fig. 13c). Secondly, we can consider the withdrawal of magma from the shallow reservoir which is inferred to exist near sea level beneath the summit (Lénat et al., 1990; Carter et al., 2007; Peltier et al., 2007). The characteristics of the eruption (huge outflow rate, presence of cumulates, large emitted volume) suggest a quasi hydrostatic draining of a magma body and the volume of the collapse matches that of the erupted magma. In this mechanism, the volume left by the magma within the edifice would be replaced by less dense fractured rocks.

6.2.2. Estimate of the mass variation

The excess (or deficit) of mass M giving rise to a gravity change can be calculated by a surface integration of the gravity variation (Grant and West, 1965):

$$M = \frac{1}{2\pi G}\iint_s \Delta g(x,y)dxdy$$

where G is the gravitational constant and Δg is the gravity change.

However, there are some practical limitations for this computation. The accuracy of the estimate of the mass change is limited by the accuracy of the observed anomaly and by the limited extent of the map, which does not cover all the extremities of the anomaly.

The surface integration of the free air anomaly difference before and after the Dolomieu collapse has been carried out on a mesh with square cells 20 m wide over a circular area of 25 km^2 centred on the summit craters (Fig. 11). The selected area covers practically the whole surface of the negative anomaly. The computation gives rise to an estimate of $853 \pm 650 \times 10^9$ kg of mass change, which would represent a volume of approximately 384 and 320×10^6 m^3 for densities of 2.2×10^3 and 2.67×10^3 kg m^{-3} respectively. Owing to the very large uncertainty, two extreme explanations can be proposed for the mass deficit between the two datasets. On the one hand, the change can be attributed exclusively to the replacement of rocks by air in the collapsed volume. On the other hand, the change may require an additional source of mass deficit amounting to 200×10^9 kg or more, with a volume ranging from 140 to 30×10^6 m^3 for densities of 2.2×10^3 to 2.67×10^3 kg m^3 respectively.

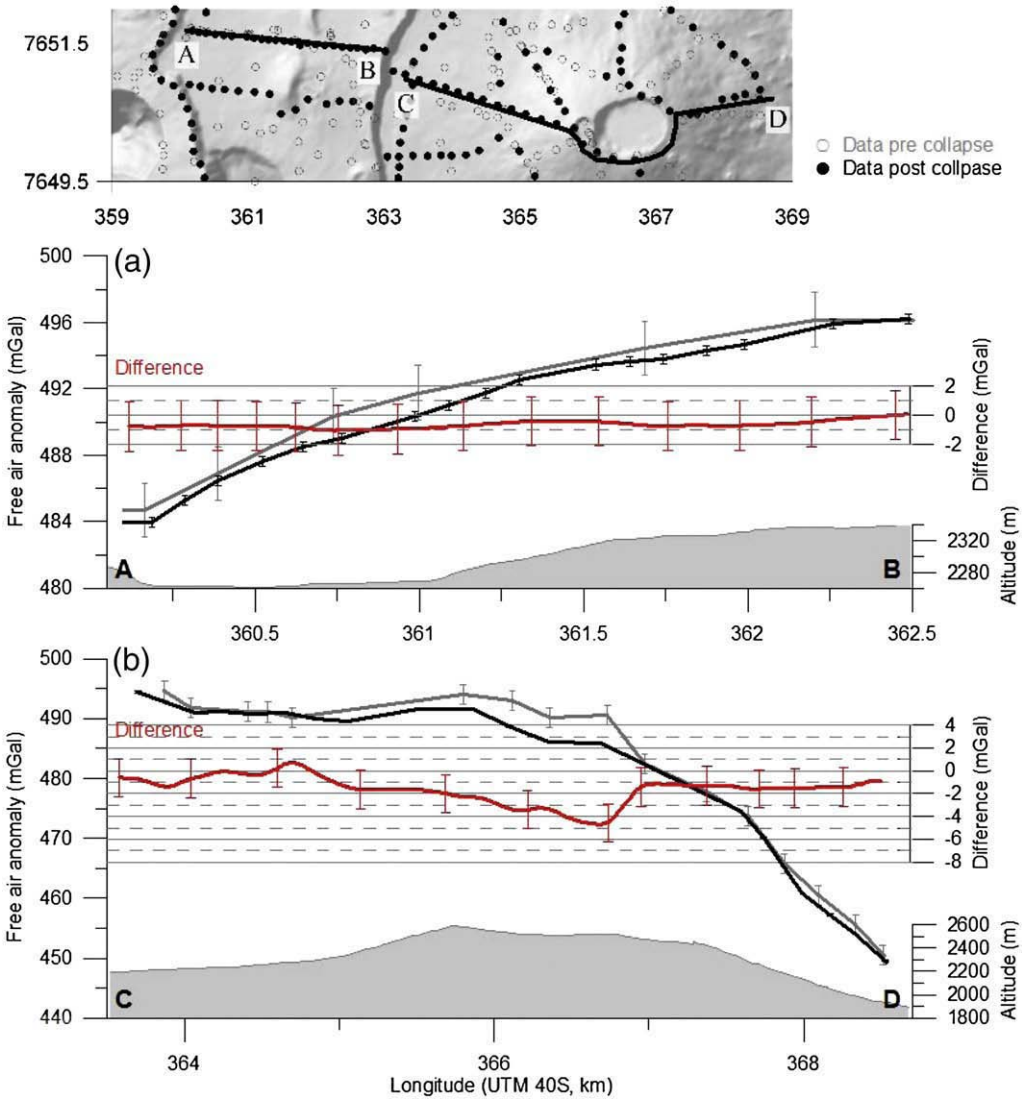

Fig. 12. Comparison of pre- and post-April 2007 gravity anomalies along profiles: (a) Plaine des Sables area; (b) Central Cone area.

6.2.3. Gravity models of structural changes

We have constructed different models corresponding to the three situations shown on Fig. 13, simulating the 3 following structures:

(1) The collapsed volume (i.e. the new Dolomieu crater; Fig. 13b). For this structure the density is inferred to be of 1.8×10^3 kg m^{-3} on the basis of the structural study described in a previous section.

(2) A magma body drained by the eruption (Fig. 13d). The 120×10^6 m^3 April 2007 lava flow, with an estimated density of 2.2×10^3 kg m^{-3}, represents a mass of 2.64×10^{11} kg. The density of basalts samples from Piton de la Fournaise (N. Villeneuve, Com. Pers.) ranges from 2.6 to 3.0×10^3 kg m^{-3}. We have thus inferred a density of 2.8×10^3 kg m^{-3} for the magma body. Considering only the lava flow and disregarding the unknown intrusion volume, the drained volume is about 94×10^6 m^3. We have assumed that it was contained in a flattened cylinder, centred beneath the Dolomieu crater, with a diameter similar to that of the crater (an estimate based on the work by Roche et al., 2001). The depth of this reservoir is derived from the analysis of the seismicity and the deformation associated with different crises (Lénat and Bachèlery, 1990; Peltier, 2007; Peltier et al., submitted for publication).

(3) The third structure is a 1 km diameter cylinder joining the bottom of Dolomieu to the drained reservoir (Fig. 13c). This structure is intended to simulate a fractured volume created by the collapse of a column between the surface and the magma reservoir. It can be inferred that the density of this column is lowered by the fracture processes, but it is difficult to assess how much. We have tentatively assumed a density decrease of 0.2×10^3 kg m^{-3}.

We have built 3D models (using GMSYS 3D software) to explore the signals generated by the three types of structures described above. Fig. 13a shows the observed anomaly and the signals of the individual structures and various combinations of the addition of the signals. If we consider only the effect of the collapsed crater (curve 1 on Fig. 13a), the resulting anomaly seems to be insufficient to explain the observed signal, both for the amplitude and for the wavelength. The influence of the cylinder-shaped reservoir at depth (curve 3 on Fig. 13a) is very low and practically negligible considering the accuracy of our data. The effect of a fractured column (curve 2 on Fig. 13a) produces a gravity response with a wavelength larger than that of the collapsed crater and with significant amplitude. The combination of the effects of the three structures (curve 5 on Fig. 13a) creates an anomaly comparable to the observed one, both in amplitude and especially in wavelength. In such a model, the individual contribution of each structure to the gravity signal is about 70% for the collapsed crater, 3% for the reservoir and 27% for the fractured column. It should be

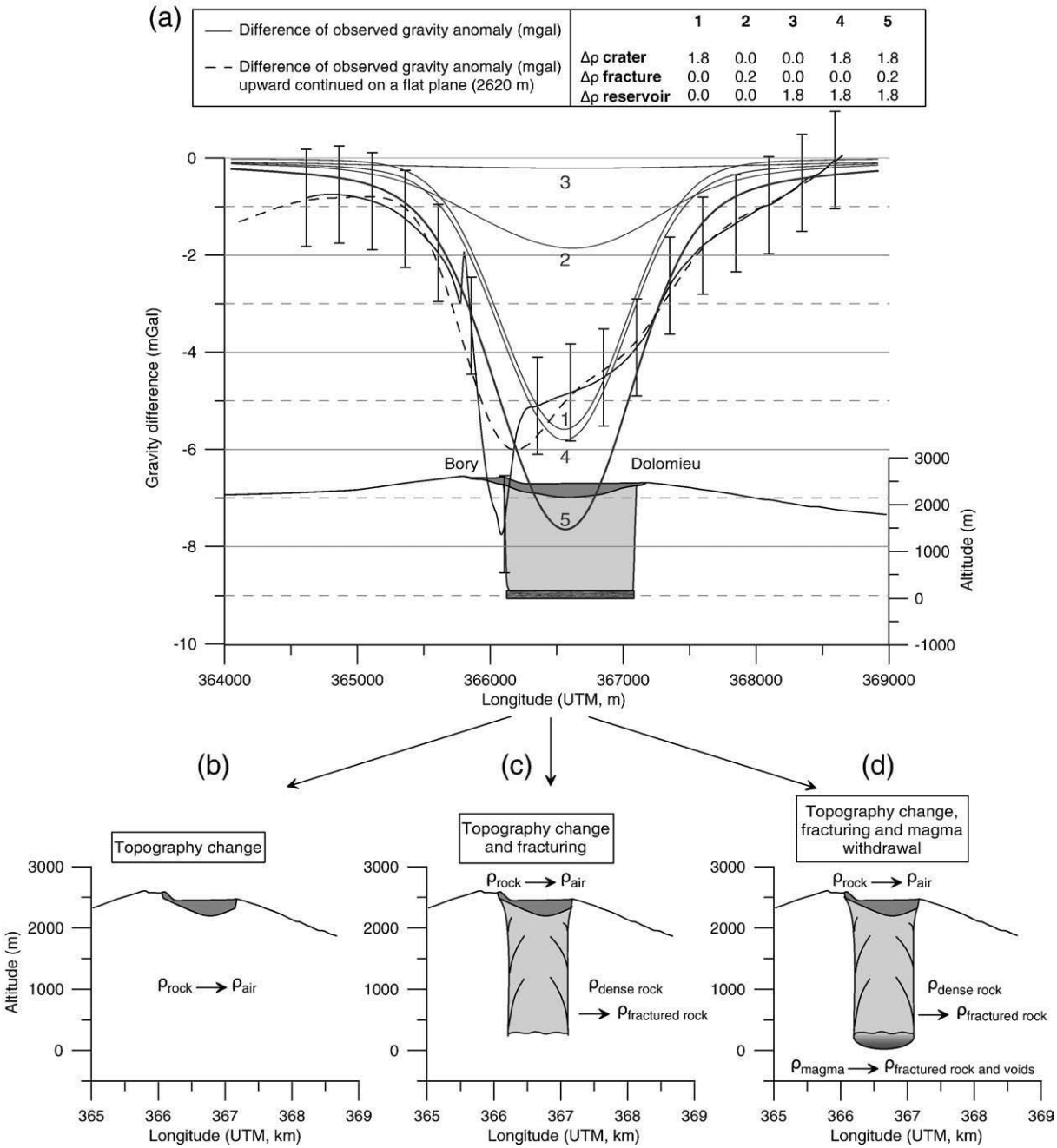

Fig. 13. Conceptual models explaining the free air anomaly decrease after the summit collapse. (a) WE profiles across 3D models (7650.25 km in latitude); the 3D models are computed on a flat plane at 2620 m asl. They show the individual and combined contributions of the different structures considered in the text. Coordinates in meters (WGS84, UTM40S). Errors bars are rms errors calculated for the free air anomaly difference. (b) to (c): possible sources for the observed density decrease.

stressed that the accuracy of the proposed model is limited by that of the data which were not intended to monitor mass transfer within the edifice. However, the models suggest that the gravity changes cannot be explained only by the new crater alone, and that internal density changes also occurred during the collapse crisis.

7. Conclusions

The gravity study presented here provides new information compared to the previously published works by Rousset et al. (1989) and Malengreau et al. (1999). The denser network of on-land and offshore gravity stations enables us to better differenciate the shallow and deeper structures of Piton de la Fournaise and to constrain the structures in the offshore continuation of the volcano. Fig. 14 summarizes the structural results of our work.

Short-wavelength positive anomalies have been attributed to piles of thick dense lava flows that infill previous volcano-tectonic depressions. On this basis, we mapped the lateral extension and the depth of paleo-depressions associated with the collapses of the Plaine des Sables-Fond de la Rivière de l'Est and the Enclos Fouqué (Fig. 14). These geological features were poorly or not at all known. They constitute new constraints to unravel the complex evolution of this area. Conversely, a negative short-wavelength anomaly is associated with the currently most active zone. This Central Cone anomaly can be explained by the nature of the rocks, now well exposed in the rim of the newly collapsed Dolomieu crater. The Central Cone is mostly composed of a pile of thin, highly vesiculated and fractured lava flows. The gravity models suggest that the cone is made of low density rocks from its base to its top. In addition, we suspect that the shallow, hydrothermally altered volume beneath the cone also contributes to

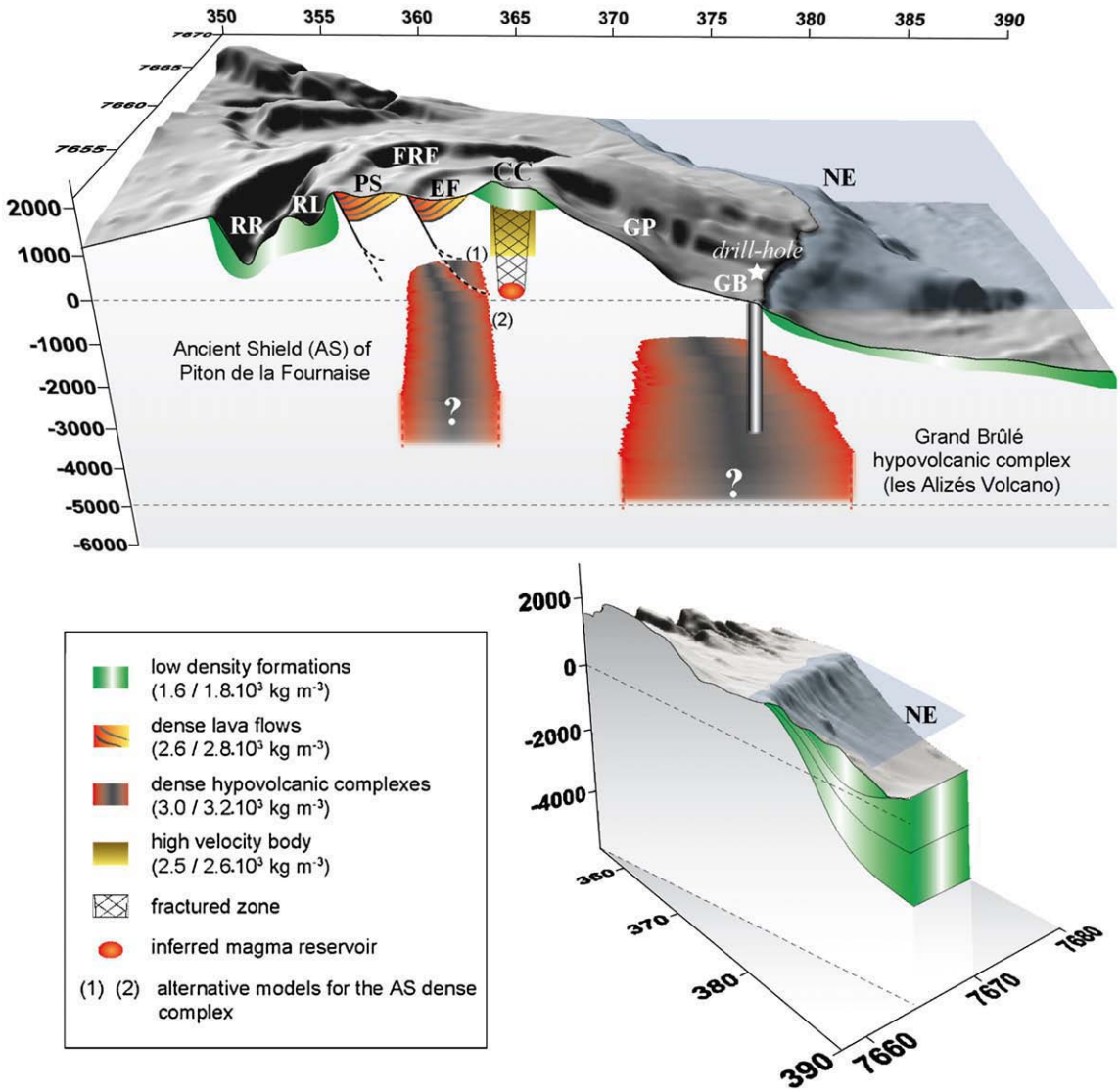

Fig. 14. Conceptual interpretative scheme of the internal structure of Piton de la Fournaise derived from gravity modelling: (a) model of the overall study area; (b) detailed of the NE rift-zone area. Labels refer to the main places discussed in text: Rivière des Remparts (RR), Rivière Langevin (RL), Plaine des Sables (PS), Fond de la Rivière de l'Est (FRE), Enclos Fouqué (EF), Central Cone (CC), Grandes Pentes (GP), Grand Brûlé (GB). NE rift zone (NE), Ancient Shield of Piton de la Fournaise (AS), Grand Brûlé (GB). Coordinates in kilometers (WGS84, UTM40S); densities in 10^3 kg m^{-3}.

the negative anomaly, as well as the column of fractured rocks between the surface collapse and a magma reservoir near sea level.

Negative short- to medium-wavelength anomalies exist in the Rivière des Remparts-Rivière Langevin area, above the offshore continuation of the NE and SE rift zones and above the Central Cone (Fig. 14). For the former zone, no explanation can be derived from the observation of outcrops in the valleys. We speculate that this zone is underlain by breccias related to erosion and/or to southward directed mass-wasting. The interpretation of the topographic highs in the offshore continuation of the NE and SE rift zones had to bring together a relatively low density and a significant thermoremanent reverse magnetization. The interpretation of these rocks as hyaloclastites provides a coherent explanation for the observed signals if we admit that they were primarily constructed during the Alizés stage.

Two main deeper, dense structures are defined (Fig. 14): the Grand Brûlé complex and a complex beneath the Plaine des Sables and the Enclos Fouqué. The Grand Brûlé structure was describes by the work of Rousset et al. (1989) and its nature as a hypovolcanic complex has been established by drilling (Rançon et al., 1989). The main contribution of the present work is to provide a map that allows us to improve definition of the geometry of the source body. The latter is clearly disconnected from the present Piton de la Fournaise volcano and its interpretation as the hypovolcanic complex of Les Alizés volcano (Lénat et al., 2001; Malengreau et al., 1999) remains valid. The most notable characteristic of this structure is its marked NS elongated geometry, a feature not observed on the other similar complexes of Plaine des Sables and Piton des Neiges (Malengreau et al., 1999). This geometry is undoubtedly inherited from an anisotropic stress field during the growth of the complex. However, no interpretation of this characteristic has been yet proposed.

The dense structure beneath the Plaine des Sables and the eastern part of the Enclos can be convincingly identified as the hypovolcanic intrusive complex of the Ancient Shield of Piton de la Fournaise. The occurrence of gabbro and peridotite xenoliths in eruptive products in this area (Bachèlery, 1981) and the seismic tomography of Hirn et al. (1999) support the presence of this underlying hypovolcanic complex. This structure is centred roughly above the Enclos fault, about 1.5 km to the east of the Ancient Shield center proposed by Bachèlery and Mairine (1990). The modelling has been inconclusive in establishing whether or not the complex is affected by the Enclos fault.

One of the difficulties encountered in the interpretation of the gravity data was an apparent contradiction between the seismic

tomographies and the gravity pattern. In particular, a high velocity cylinder is present on all the published tomographies beneath the summit, extending virtually from the surface to at least 1.5 km in depth. In the same area we observe a gravity low. Since high seismic velocities usually correspond to high densities values (Gebrande et al., 1982), the two results seemed incompatible. This apparent contradiction is resolved if we consider the relative sensitivity of each method. Strong shallow gravity signals, such as the one produced by the low dense Central Cone, will significantly hide a moderate signal from a deeper, denser structure (Fig. 14). Conversely, with a moderate station and signal coverage, the seismic tomographies will fail to define precisely the subsurface structures. The combination of both methods increases the accuracy in determining the internal structure of the central area.

Finally, this gravity study has permitted us to go beyond the static structure of Piton de la Fournaise and to explore the gravity changes that accompanied the major eruptive and volcano-tectonic crisis of April 2007. A 350 m collapse of the summit and the emission of $\sim 120 \times 10^6$ m^3 of lava created a change in the mass distribution of the edifice large enough to be detected in the data collected in the central area before and after the crisis. However, the gravity change was determined with too low accuracy for the precise assessment of the mass transfers. The mass deficit observed between before and after the April 2007 crisis can be explained by the creation of the new Dolomieu crater only, or by the sum of the effects of the crater formation, the fracturing of a column of rocks between the surface and the drained magma reservoir and the draining of the magma reservoir (Fig. 14).

Acknowledgements

Financial support for the 2001–07 surveys was provided by the Conseil Régional de La Réunion (Geothermal Project) and the VOLCARISK ANR. The gravimeters were provided by the "Parc Instrumental" from INSU. We would like to address special thanks to Jean-François Oehler, Bernard Contarin and Pierre Tinard for their help during the gravity field survey. We wish also to thank the captains and crew of the R/V Beautemps Beaupré for their efficient work at sea during the ERODER 1 marine survey and the scientific leaders of this cruise, Bruno Savoy and Patrick Bachèlery. We express our sorrow at the tragic death of Bruno Savoy during summer 2008. Finally, we thank the staffs of the Observatoire Volcanologique du Piton de la Fournaise and of Université de la Réunion for their frequent logistic help during fieldwork. The paper benefited from helpful reviews and comments by Jim Kauahikaua, and an anonymous reviewer.

References

Bachèlery, P., 1981. Le Piton de la Fournaise (Ile de la Reunion). Etude volcanologique, structurale et petrologique. Piton de la Fournaise, Reunion; volcanologic, structural and petrographic study. Doctoral Thesis, Univ. of Clermont-Ferrand, Clermont-Ferrand, 255 pp.

Bachèlery, P., 1995. Quelques réflexions à propos de concepts récents sur la structure du Piton de la Fournaise, Réunion. Rapport Quadriennal 1991–1994. Comité National Français de Géodésie et de Géophysique, Paris.

Bachèlery, P., Chevallier, L., 1982. Carte volcano-structurale du massif du Piton de la Fournaise au 1:50000 ème avec notice explicative. Institut de Physique du Globe de Paris.

Bachèlery, P., Lénat, J.-F., 1993. Le Piton de la Fournaise. Mémoires de la Soc. Géol. France 163, 221–229 Nouvelle Série.

Bachèlery, P., Mairine, P., 1990. Evolution volcano-structurale du Piton de la Fournaise depuis 0.53 Ma. In: Lénat, J.-F. (Ed.), Le volcanisme de la Réunion, Monographie. Cent. Rech. Volcanol., Clermont-Ferrand, France, pp. 213–242.

Brenguier, F., Shapiro, N.M., Campillo, M., Nercessian, A., Ferrazzini, V., 2007. 3-D surface wave tomography of the Piton de la Fournaise volcano using seismic noise correlations. Geophys. Res. Lett. 34, L02305.

Brown, G.C., Rymer, H., Stevenson, D., 1991. Volcano monitoring by microgravity and energy budget analysis. J. Geol.l Soc. 148 (3), 585–593.

Carter, B., van Wyk de Vries, B., Kelfoun, K., Bachèlery, P., Briole, P., 2007. Pits, rifts and slumps: the summit structure of Piton de la Fournaise. Bull. Volcanol. 69 (7), 741–756.

Chevallier, L., Bachèlery, P., 1981. Evolution structurale du volcan actif du Piton de la Fournaise. Ile de la Réunion, Océan Indien. Bull. Volcanol. 44 (4), 723–741.

Courteaud, M., 1996. Etude des structures géologiques et hydrogéologiques du massif de la Fournaise par la méthode audiomagnétotellurique. Université de la Réunion. 212 pp.

de Voogd, B., Palome, S., Hirn, A., Charvis, P., Gallart, J., Rousset, D., Danobeitia, J., Perroud, H., 1999. Vertical movements and material transport during hotspot activity; seismic reflection profiling offshore La Reunion. J. Geophys. Res. B104 (2), 2855–2874.

Delorme, H., Bachèlery, P., Blum, P.A., Cheminée, J.-L., Delarue, J.F., Delmond, J.C., Hirn, A., Lépine, J.C., Vincent, P.M., Zlotnicki, J., McBirney, A.R., 1989. March 1986 eruptive episodes at Piton de la Fournaise Volcano (Reunion Island). J. Volcanol. Geotherm. Res. 36 (1–3), 199–208.

Demange, H., Chovelon, J., Puvilland, O., 1989. Geothermal model of the Salazie Cirque (Réunion Island): volcanic and structural implications. J. Volcanol. Geotherm. Res. 36, 153–176.

Deniel, C., Kieffer, G., Lecointre, J., 1992. New ^{230}Th–^{238}U and ^{14}C age determinations from Piton des Neiges volcano, Réunion — a revised chronology for the differentiated series. J. Volcanol Geotherm. Res. 51, 253–267.

Duclaux, F., Martin, J., Blot, C., Remiot, R., 1954. Etablissement d'un réseau général de stations gravimétriques en Afrique, à Madagascar, à la Réunion et à l'île Maurice. ORSTOM, Paris. 1–223 pp.

Duncan, R.A., Backman, J., Peterson, L., 1989. Reunion hotspot activity through tertiary time: initial results from the Ocean Drilling Program, Leg 115. J. Volcanol. Geotherm. Res. 36, 193–198.

Famin, V.,Welsh, B., Okumura, S., Bachèlery, P., Nakashima, S., 2009. Three differentiation stages of a single magma at Piton de la Fournaise (Reunion hotspot). G-Cubed. 10 (1) Q01007. doi:10.1029/2008GC002015.

Gebrande, H., Kern, H., Rummel, F., 1982. Landolt-Börnstein numerical data and functional relationship in science and technology. Group V. Geophysics and Space Research, Physical Properties of Rocks, Subvolume b, 1. Springer-Verlag, Berlin. 1–223 pp.

Gérard, A., Lesquer, A., Lachaud, J.-C., Louis, P., Mennechet, C., 1980. Etude gravimétrique de la moitié sud-est de l'île de la Réunion. C.R. Acad. Sci. Paris 290, 139–142.

Gillot, P.-Y., Lefèvre, J.-C., Nativel, P.-E., 1994. Model for the structural evolution of the volcanoes of Réunion. Earth Planet. Sci. Lett. 122 (291–302).

Gillot, P.-Y., Nativel, P., 1982. K-Ar chronology of the ultimate activity of Piton des Neiges volcano, Réunion. J. Volcanol. Geotherm. Res. 13, 131–146.

Gillot, P.-Y., Nativel, P., 1989. Eruptive history of the Piton de la Fournaise volcano, Réunion Island, Indian Ocean. J. Volcanol. Geotherm. Res. 36 (53–65).

Grant, F.S., West, G.F., 1965. Interpretation Theory in Applied Geophysics. McGraw-Hill Book Co, New York. 584 pp.

Harrison, G.G.A., Ball, M.M., 1974. Geophysical observations on an exposed seamount in the Afar depression. Bull. Volcanol. 38 (1), 1–18.

Hirn, A., Lépine, J., Sapin, M., Laigle, M., Nicolich, R., Gallart, J., Lankar, V., Nercessian, A., De Voogd, B., Charvis, P., Veinante, J., Verhille, J., 1999. Structure interne du Piton de la Fournaise et de l'Etna d'après la tomographie sismique. Rapport quadriennal 95–98 du CNFGG. Comité National Français de Géodésie et de Géophysique, Paris, pp. 129–136.

Kauahikaua, J., Hildenbrand, T., Webring, M., 2000. Deep magmatic structures of Hawaiian volcanoes, imaged by three-dimensional gravity models. Geology 10883–886, 883–886.

Labazuy, P., 1991. Instabilités au cours de l'évolution d'un édifice volcanique en domaine intraplaque océanique: le Piton de la Fournaise (Ile de la Réunion). Approche pluridisciplinaire à partir des données de campagnes marines, Univ. Blaise Pascal, Clermont-Ferrand.

Labazuy, P., 1996. Recurrent landslides events on the submarine flank of Piton de la Fournaise volcano (Réunion Island). In: McGuire, W., Jones, A.P., Neuberg, J. (Eds.), Volcano Instability on the Earth and other Planets. Geological Society, London, pp. 293–305.

Labazuy, P., Lénat, J.-F., 1990. Recurrent landslides on the east flank of Piton de la Fournaise Volcano, Reunion. Eos, Trans., Am. Geophys. Union 71 (43), 1577.

Lacroix, A., 1936. Le volcan actif de la Réunion et ses produits. Gauthier-Villars, Paris. 295 pp.

Lacroix, A., 1938. Le volcan actif de la Réunion (supplément) et celui de la Grande Comore. Gauthier-Villars, Paris. 57 pp.

Lafehr, T.-R., 1991b. An exact solution for the gravity curvature (Bullard B) correction. Geophysics 56 (8), 1179–1184.

Lambert, M., 2003. Etude de la structure interne du point chaud de la Réunion d'après les données gravimétriques. Université Blaise Pascal-CNRS, Clermont-Ferrand. pp. 52.

Lénat, J.-F., Bachèlery, P., 1990. Structure et fonctionnement de la zone centrale du Piton de la Fournaise. In: Lénat, J.-F. (Ed.), Le Volcanisme de La Réunion Monographie. Centre de RecherchesVolcanologiques. Clermont Ferrand, France, pp. 257–296.

Lénat, J.-F., Bachèlery, P., Galdéano, A., Labazuy, P., Rousset, D., Vincent, P., 1990. Structure and morphology of the submarine flank of an active basaltic volcano: Piton de la Fournaise (Réunion Island, Indian Ocean). Oceanologica Acta 10, 211–223.

Lénat, J.-F., Fitterman, D., Jackson, D.B., Labazuy, P., 2000. Geoelectrical structure of the central zone of Piton de la Fournaise volcano (Réunion). Bulletin of Volcanology 62, 75–89.

Lénat, J.-F., Gibert-Malengreau, B., Galdeano, A., 2001. A new model for the evolution of the volcanic island of Reunion (Indian Ocean). J. Geophys. Res., B 106 (5), 8645–8663.

Lesquer, A., 1990. Structure profonde de l'île de la Réunion d'aprés l'étude des anomalies gravimétriques. In: Lénat, J.-F. (Ed.), Le volcanisme de la Réunion, Monographie. Cent. Rech. Volcanol., Clermont-Ferrand, France, pp. 19–27.

Levieux, G., 2004. Synthèse géophysique de la zone de forage de reconnaissance géothermique du Piton de la Fournaise. Université Blaise Pascal, Clermont-Ferrand, p. 65.

Li, Y., Oldenburg, D.W., 1998. 3D inversion of gravity data. Geophysics 63, 361–371.
Longpré, M.-A., Staudacher, T., Stix, J., 2007. The November 2002 eruption at Piton de la Fournaise volcano, La Réunion Island: Ground deformation, seismicity, and pit crater collapse. Bulletin of Volcanology 69, 511–525.
Malengreau, B., 1995. Structure profonde de La Réunion d'après les données magnétiques et gravimétriques, Blaise Pascal, Clermont-Ferrand, France.
Malengreau, B., Lénat, J.-F., Froger, J.-L., 1999. Structure of Reunion Island (Indian Ocean) inferred from the interpretation of gravity anomalies. Journal of Volcanology and Geothermal Research 88 (3), 131–146.
Merle, O., Lénat, J.-F., 2003. Hybrid collapse mechanism at Piton de la Fournaise volcano, Reunion Island, Indian Ocean. J. Geophys. Res. 108 (B3), 2166.
Michon, L., Staudacher, T., Ferrazzini, V., Bachelery, P., Marti, J., 2007. April 2007 collapse of Piton de la Fournaise: a new example of caldera formation. Geophys. Res. Lett. 34, L21301.
Moore, J.G., 2001. Density of basalt core from Hilo drill hole, Hawaii. J. Volcanol. Geotherm. Res. 112 (1–4), 221–230.
Nercessian, A., Hirn, A., Lépine, J.-C., Sapin, M., 1996. Internal structure of Piton de la Fournaise volcano from seismic wave propagation and earthquake distribution. J. Volcanol. Geotherm. Res. 70, 123–143.
Nettleton, L.-C., 1939. Determination of density of reduction of gravimeter observation. Geophysics 4, 176–183.
Oehler, J.-F., Lénat, J.-F., Labazuy, P., 2007. Growth and collapse of the Reunion Island volcanoes. Bull. Volcanol. doi:10.1007/s00445-007-0163-0.
Peltier, A., 2007. Suivi, Modélisation et Evolution des processus d'injections magmatiques au Piton de La Fournaise. Doctorat d'Université Thesis, Université de La Réunion, 365 pp.
Peltier, A., Staudacher, T., Bachèlery, P., 2007. Constraints on magma transfers and structures involved in the 2003 activity at Piton de La Fournaise from displacement data. J. Geophys. Res. 112 (B03207).
Peltier, A., Bachèlery, P. and Staudacher, T., submitted for publication. Magma transfer and storage at Piton de La Fournaise (La Réunion Island): a review of 35 years of eruptive activity from geophysical and geochemical data. Earth Science Reviews, submitted.
Prôno, E., Battaglia, J., Monteiller, V., Got, J.-L., Ferrazzini, V., 2009. P-wave velocity structure of Piton de la Fournaise volcano deduced from seismic data recorded between 1996 and 1999. J. Volcanol. Geotherm. Res. 184, 49–62 (this issue).
Rançon, J.-P., Lerebour, P., Augé, T., 1989. The Grand Brule exploration drilling; new data on the deep framework of the Piton de la Fournaise Volcano; Part 1, Lithostratigraphic units and volcanostructural implications. J. Volcanol. Geotherm. Res. 36 (1–3), 113–127.
Rechenmann, J., 1976. Madagascar et autres îles du sud-ouest de l'Océna Indien (cartes des anomalies de Bouguer feuille sud). ORSTOM, Paris.
Roche, O., van Wyk de Vries, B., Druitt, T.H., 2001. Sub-surface structures and collapse mechanisms o f summit pit craters. J. Volcanol. Geotherm. Res. 105, 1–18.
Rousset, D., Bonneville, A., Lénat, J.-F., 1987. Detailed gravity study of the offshore structure of Piton de la Fournaise volcano, Réunion Island. Bull.Volcanol. 49, 713–722.
Rousset, D., Lesquer, A., Bonneville, A., Lenat, J.F., McBirney, A.R., 1989. Complete gravity study of Piton de la Fournaise Volcano, Reunion Island. J. Volcanol. Geotherm. Res. 36 (1–3), 37–52.
Rymer, H., 1989. A contribution to precision microgravity data analysis using LaCoste and Romberg gravity meters. Geophys. J. 97, 311–322.
Urai, M., Geshi, N., Staudacher, T., 2007. Size and volume evaluation of the caldera collapse on Piton de la Fournaise volcano during the April 2007 eruption using ASTER stereo imagery. Geophys. Res. Lett. 34, L22318.